

Estero Bay Aquatic Preserve

SEACAR Habitat Analyses

Last compiled on 10 June, 2026

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Funding & Acknowledgements

The data used in this analysis is from the Export Standardized Tables in the SEACAR Data Discovery Interface (DDI). Documents and information available through the SEACAR DDI are owned by the data provider(s) and users are expected to provide appropriate credit following accepted citation formats. Users are encouraged to access data to maximize utilization of gained knowledge, reducing redundant research and facilitating partnerships and scientific innovation.

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Threshold Filtering

Threshold filters, following the guidance of Florida Department of Environmental Protection's (*FDEP*) Division of Environmental Assessment and Restoration (*DEAR*) are used to exclude specific results values from the SEACAR Analysis. Based on the threshold filters, Quality Assurance / Quality Control (*QAQC*) Flags are inserted into the *SEACAR_QAQCFlagCode* and *SEACAR_QAQC_Description* columns of the export data. The *Include* column indicates whether the *QAQC* Flag will also indicate that data are excluded from analysis. No data are excluded from the data export, but the analysis scripts can use the *Include* column to exclude data (1 to include, 0 to exclude).

Table 1: Continuous Water Quality threshold values

<i>Parameter Name</i>	<i>Units</i>	<i>Low Threshold</i>	<i>High Threshold</i>
Chlorophyll a, Uncorrected for Pheophytin	ug/L	-	-
Dissolved Oxygen	mg/L	-0.000001	50
Dissolved Oxygen Saturation	%	-0.000001	500
Fluorescent Dissolved Organic Matter	QSE	-	-
Salinity	ppt	-0.000001	70
Specific Conductivity	mS/cm	-0.000001	200
Turbidity	NTU	-0.000001	4000
Water Temperature	Degrees C	-5.000000	45
pH	None	2.000000	14

Table 2: Discrete Water Quality threshold values

<i>Parameter Name</i>	<i>Units</i>	<i>Low Threshold</i>	<i>High Threshold</i>
Ammonia, Un-ionized (NH3)	mg/L	-	-
Ammonium (NH4)	mg/L	-	-

<i>Parameter Name</i>	<i>Units</i>	<i>Low Threshold</i>	<i>High Threshold</i>
Chlorophyll a, Corrected for Pheophytin	ug/L	-	-
Chlorophyll a, Uncorrected for Pheophytin	ug/L	-	-
Colored Dissolved Organic Matter	PCU	-	-
Dissolved Oxygen	mg/L	-0.000001	25
Dissolved Oxygen Saturation	%	-0.000001	310
Fluorescent Dissolved Organic Matter	QSE	-	-
Light Extinction Coefficient	m ⁻¹	-	-
NO2+3, Filtered	mg/L	-	-
Nitrate (NO3)	mg/L	-	-
Nitrite (NO2)	mg/L	-	-
Nitrogen, inorganic	mg/L	-	-
Nitrogen, organic	mg/L	-	-
Phosphate, Filtered (PO4)	mg/L	-	-
Salinity	ppt	-0.000001	70
Secchi Depth	m	0.000001	50
Specific Conductivity	mS/cm	0.005000	100
Total Ammonia (N)	mg/L	-	-
Total Kjeldahl Nitrogen	mg/L	-	-
Total Nitrogen	mg/L	-	-
Total Nitrogen	mg/L	-	-
Total Phosphorus	mg/L	-	-
Total Suspended Solids	mg/L	-	-
Turbidity	NTU	-	-
Water Temperature	Degrees C	3.000000	40
pH	None	2.000000	13

Table 3: Quality Assurance Flags inserted based on threshold checks listed in Table 1 and 2

<i>SEACAR QAQC Description</i>	<i>Include</i>	<i>SEACAR QAQCFlagCode</i>
Exceeds maximum threshold	0	2Q
Below minimum threshold	0	4Q
Within threshold tolerance	1	6Q
No defined thresholds for this parameter	1	7Q

Value Qualifiers

Value qualifier codes included within the data are used to exclude certain results from the analysis. The data are retained in the data export files, but the analysis uses the *Include* column to filter the results.

STORET and WIN value qualifier codes

Value qualifier codes from *STORET* and *WIN* data are examined with the database and used to populate the *Include* column in data exports.

Table 4: Value Qualifier codes excluded from analysis

<i>Qualifier Source</i>	<i>Value Qualifier</i>	<i>Include</i>	<i>MDL</i>	<i>Description</i>
STORET-WIN	H	0	0	Value based on field kit determination; results may not be accurate
STORET-WIN	J	0	0	Estimated value
STORET-WIN	V	0	0	Analyte was detected at or above method detection limit
STORET-WIN	Y	0	0	Lab analysis from an improperly preserved sample; data may be inaccurate

Discrete Water Quality Value Qualifiers

The following value qualifiers are highlighted in the Discrete Water Quality section of this report. An exception is made for **Program 476** - *Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network* and data flagged with Value Qualifier **H** are included for this program only.

H - Value based on field kit determination; results may not be accurate. This code shall be used if a field screening test (e.g., field gas chromatograph data, immunoassay, or vendor-supplied field kit) was used to generate the value and the field kit or method has not been recognized by the Department as equivalent to laboratory methods.

I - The reported value is greater than or equal to the laboratory method detection limit but less than the laboratory practical quantitation limit.

Q - Sample held beyond the accepted holding time. This code shall be used if the value is derived from a sample that was prepared or analyzed after the approved holding time restrictions for sample preparation or analysis.

S - Secchi disk visible to bottom of waterbody. The value reported is the depth of the waterbody at the location of the Secchi disk measurement.

U - Indicates that the compound was analyzed for but not detected. This symbol shall be used to indicate that the specified component was not detected. The value associated with the qualifier shall be the laboratory method detection limit. Unless requested by the client, less than the method detection limit values shall not be reported

Systemwide Monitoring Program (SWMP) value qualifier codes

Value qualifier codes from the *SWMP* continuous program are examined with the database and used to populate the *Include* column in data exports. *SWMP* Qualifier Codes are indicated by *QualifierSource=SWMP*.

Table 5: SWMP Value Qualifier codes

<i>Qualifier Source</i>	<i>Value Qualifier</i>	<i>Include</i>	<i>Description</i>
SWMP	-1	1	Optional parameter not collected
SWMP	-2	0	Missing data
SWMP	-3	0	Data rejected due to QA/QC
SWMP	-4	0	Outside low sensor range
SWMP	-5	0	Outside high sensor range
SWMP	0	1	Passed initial QA/QC checks
SWMP	1	0	Suspect data
SWMP	2	1	Reserved for future use
SWMP	3	1	Calculated data: non-vented depth/level sensor correction for changes in barometric pressure
SWMP	4	1	Historical: Pre-auto QA/QC
SWMP	5	1	Corrected data

Water Column

The water column habitat extends from the water's surface to the bottom sediments, and it's where fish, dolphins, crabs and people swim! So much life makes its home in the water column that the health of marine and coastal ecosystems, as well as human economies, depend on the condition of this vulnerable habitat. Local patterns of rainfall, temperature, winds and currents can rapidly change the condition of the water column, while global influences such as [El Niño/La Niña](#), large-scale fluctuation in sea temperatures and climate change can have long-term effects. Inputs from the prosperity of our day-to-day lives including farming, mining and forestry, and emissions from power generation, automobiles and water treatment can also alter the health of the water column. Acting alone or together, each input can have complex and lasting effects on habitats and ecosystems.

SEACAR evaluates water column health with several essential parameters. These include nutrient surveys of nitrogen and phosphorus, and water quality assessments of salinity, dissolved oxygen, pH, and water temperature. Water clarity is evaluated with Secchi depth, turbidity, levels of chlorophyll a, total suspended solids, and colored dissolved organic matter. Additionally, the richness of nekton is indicated by the abundance of free-swimming fishes and macroinvertebrates like crabs and shrimps.

Seasonal Kendall-Tau Analysis

Indicators must have a minimum of five to ten years, depending on the habitat, of data within the geographic range of the analysis to be included in the analysis. Ten years of data are required for discrete parameters, and five years of data are required for continuous parameters. If there are insufficient years of data, the number of years of data available will be noted and labeled as "insufficient data to conduct analysis". Further, for the preferred Seasonal Kendall-Tau test, there must be data from at least two months in common across at least two consecutive years within the RCP managed area being analyzed. Values that pass both of these tests will be included in the analysis and be labeled as *Use_In_Analysis* = **TRUE**. Any that fail either test will be excluded from the analyses and labeled as *Use_In_Analysis* = **FALSE**. The points for all Water Column plots displayed in this section are monthly averages. Trend significance will be denoted as "Significant Trend" (when $p < 0.05$), or "Non-significant Trend" (when $p \geq 0.05$). Any parameters with insufficient data to perform Seasonal Kendall-Tau test will have their monthly averages plotted without a corresponding trend line.

Water Quality - Discrete

The following files were used in the discrete analysis:

- *Combined_WQ_WC_NUT_Chlorophyll_a_corrected_for_pheophytin-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Chlorophyll_a_uncorrected_for_pheophytin-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Colored_dissolved_organic_matter_CDOM-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Dissolved_Oxygen-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Dissolved_Oxygen_Saturation-2026-May-18.txt*
- *Combined_WQ_WC_NUT_pH-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Salinity-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Secchi_Depth-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Nitrogen-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Phosphorus-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Suspended_Solids_TSS-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Turbidity-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Water_Temperature-2026-May-18.txt*

Chlorophyll a, Corrected for Pheophytin - Discrete

Seasonal Kendall-Tau Trend Analysis

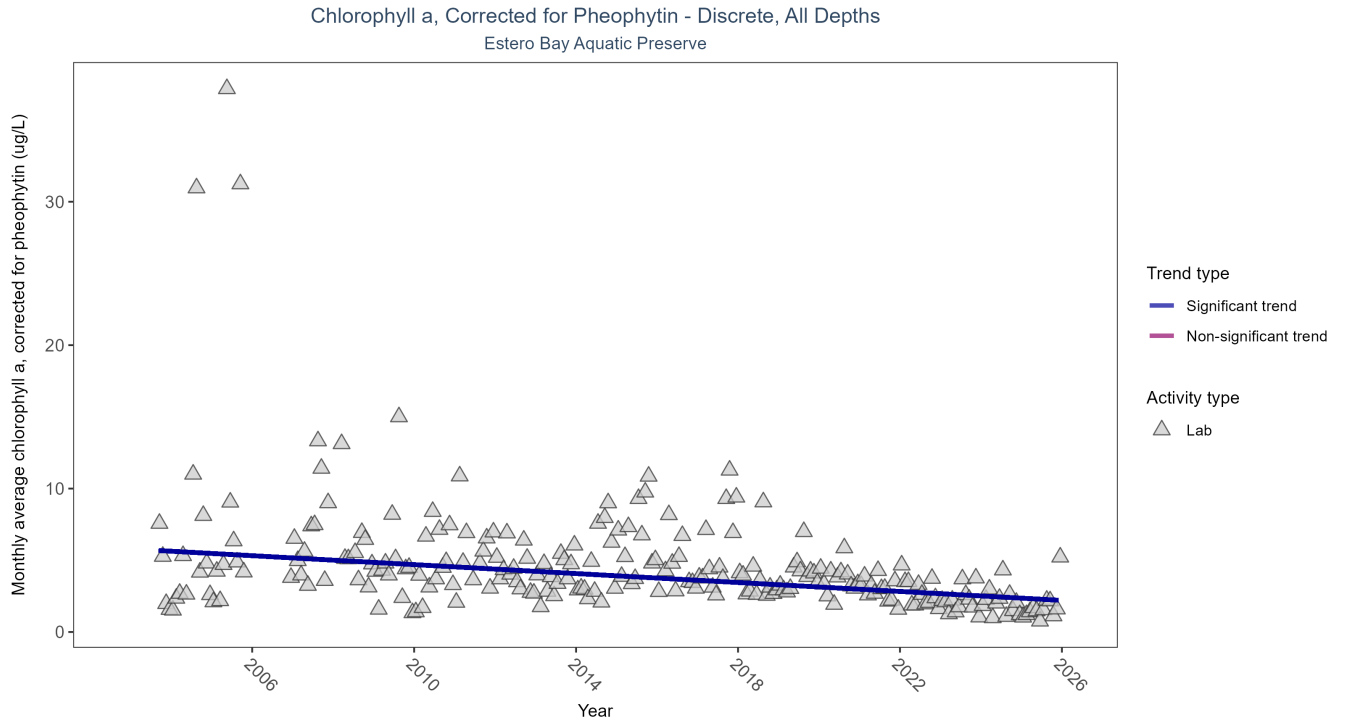


Figure 1: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 6: Seasonal Kendall-Tau Trend Analysis for Chlorophyll a, Corrected for Pheophytin

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	2673	23	2003 - 2025	2.5	-0.346	5.7883	-0.1558	0

Monthly average chlorophyll a, corrected for pheophytin, decreased by 0.16 $\mu\text{g/L}$ per year, indicating an increase in water clarity.

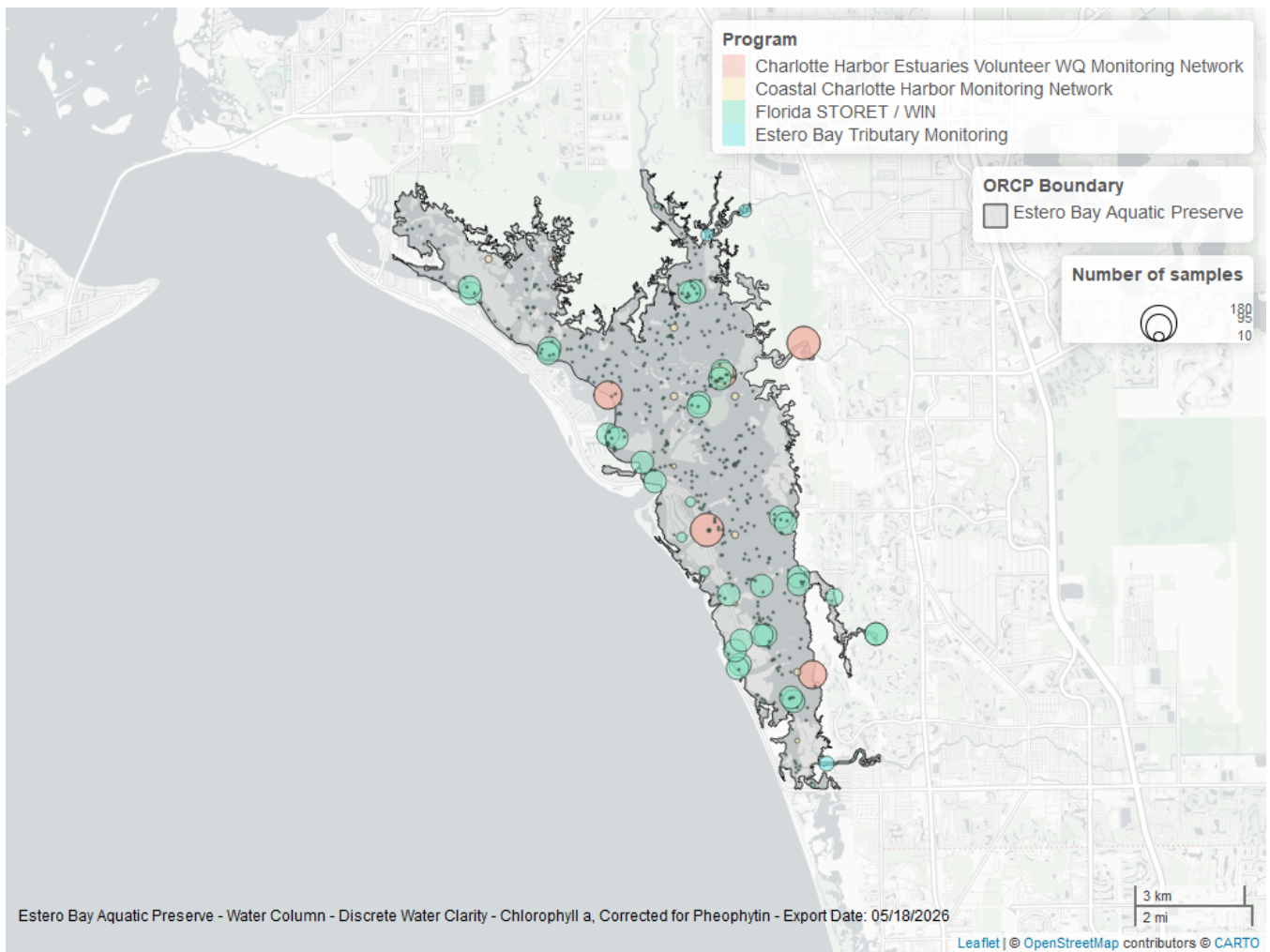


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Estero Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 7: Programs contributing data for Chlorophyll a, Corrected for Pheophytin

ProgramID	N_Data	YearMin	YearMax
5002	1877	2006	2025
476	687	2008	2025
4063	91	2018	2025
513	80	2003	2005

Program names:

5002 - Florida STORET / WIN¹

476 - Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network²

4063 - Estero Bay Tributary Monitoring³

513 - Coastal Charlotte Harbor Monitoring Network⁴

Chlorophyll a, Uncorrected for Pheophytin - Discrete

Seasonal Kendall-Tau Trend Analysis

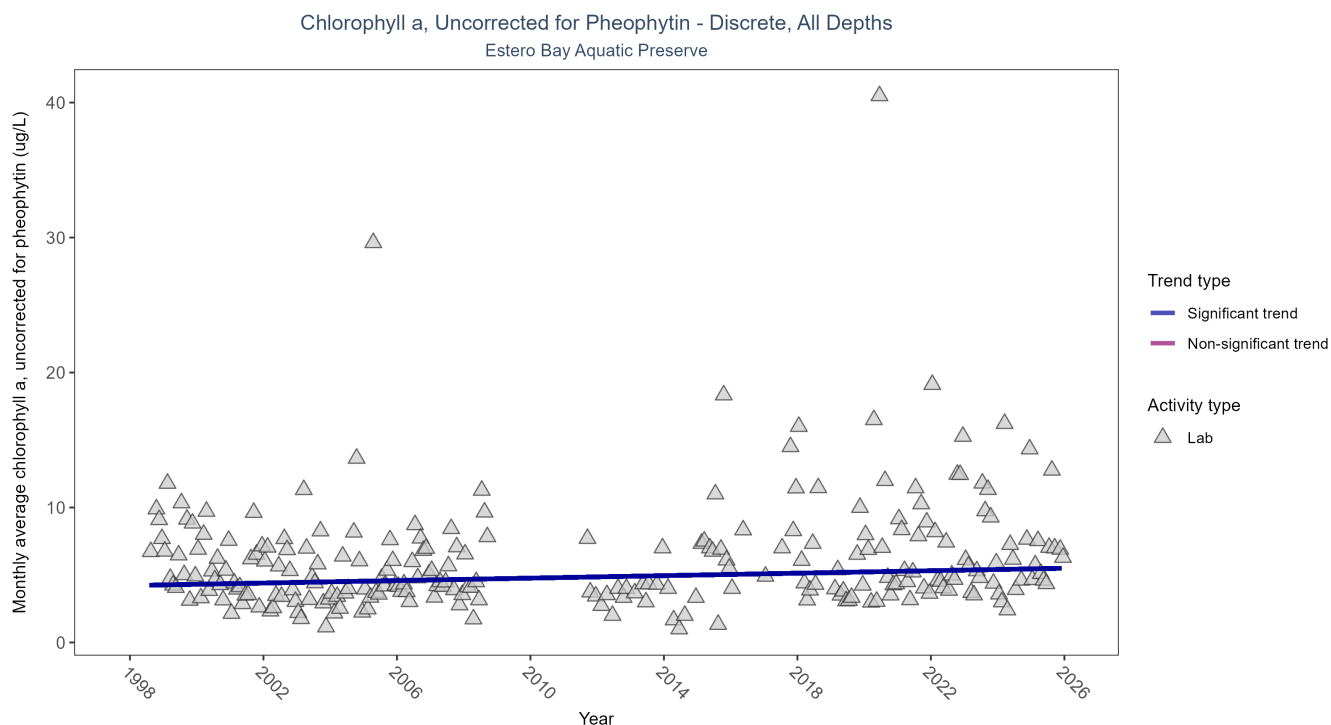


Figure 3: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 8: Seasonal Kendall-Tau Trend Analysis for Chlorophyll a, Uncorrected for Pheophytin

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	1186	26	1998 - 2025	4.4	0.1291	4.2194	0.0457	0.0047

Monthly average chlorophyll a, uncorrected for pheophytin, increased by 0.05 $\mu\text{g/L}$ per year, indicating a decrease in water clarity.

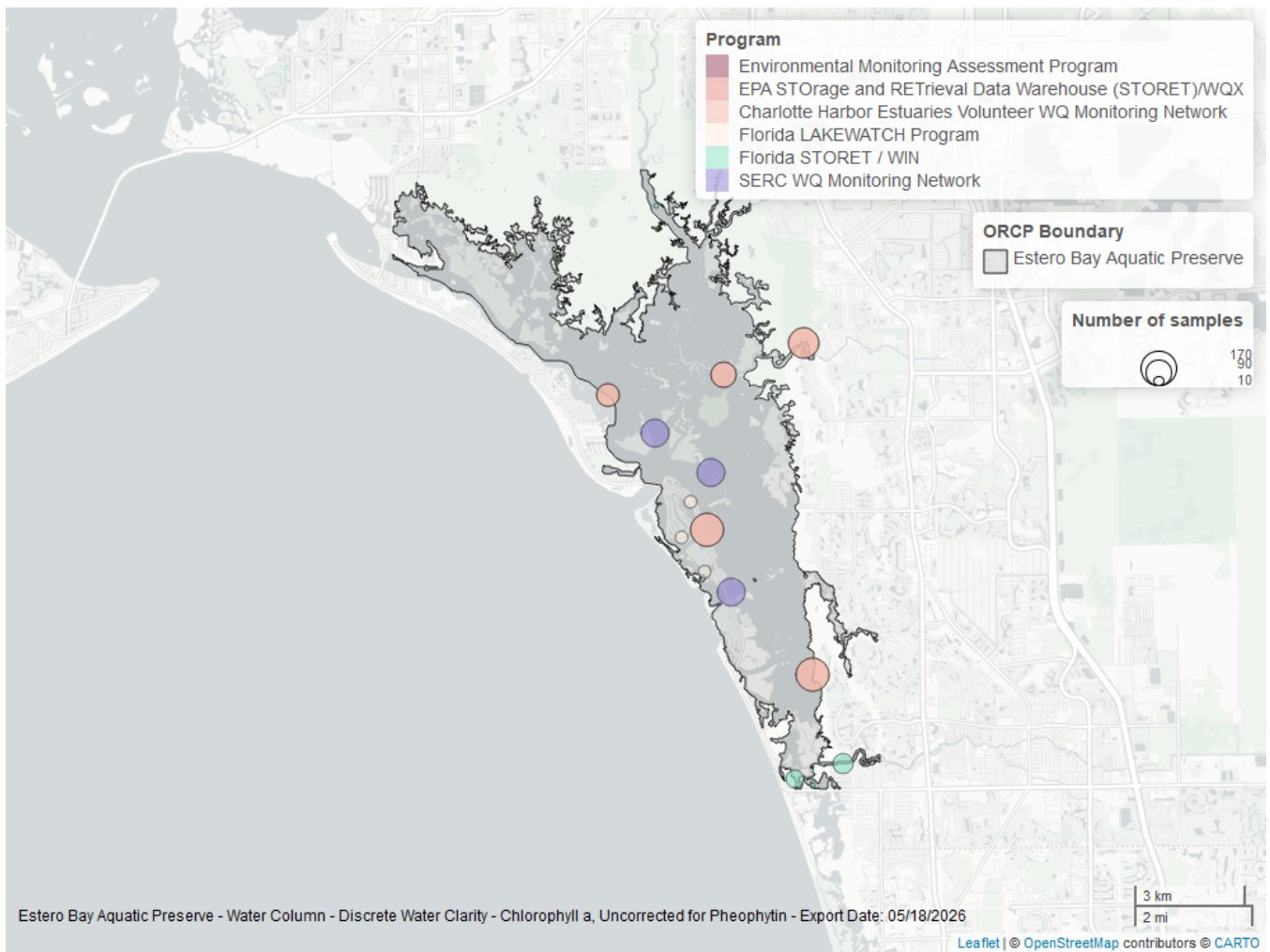


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Estero Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 9: Programs contributing data for Chlorophyll a, Uncorrected for Pheophytin

ProgramID	N_Data	YearMin	YearMax
476	645	1998	2025
509	347	1999	2008
5002	127	2011	2025
514	82	2011	2018
115	1	2003	2003
103	1	2003	2003

Program names:

[476](#) - Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network²

[509](#) - SERC Water Quality Monitoring Network⁵

[5002](#) - Florida STORET / WIN¹

[514](#) - Florida LAKEWATCH Program⁶

[115](#) - Environmental Monitoring Assessment Program⁷

[103](#) - EPA STORage and RETrieval Data Warehouse (STORET)/WQX⁸

Colored Dissolved Organic Matter - Discrete

Seasonal Kendall-Tau Trend Analysis

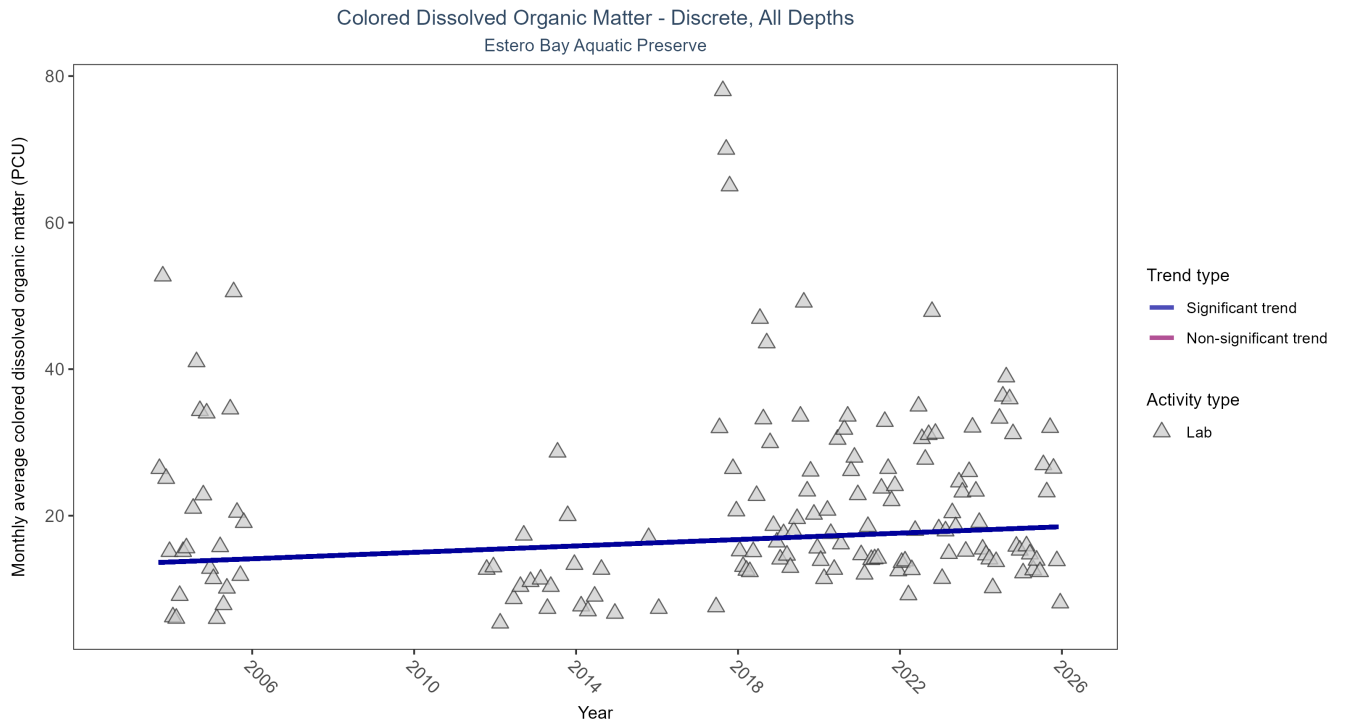


Figure 5: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 10: Seasonal Kendall-Tau Trend Analysis for Colored Dissolved Organic Matter

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	2288	18	2003 - 2025	14	0.1517	13.4626	0.2189	0.015

Monthly average colored dissolved organic matter increased by 0.22 PCU per year, indicating a decrease in water clarity.

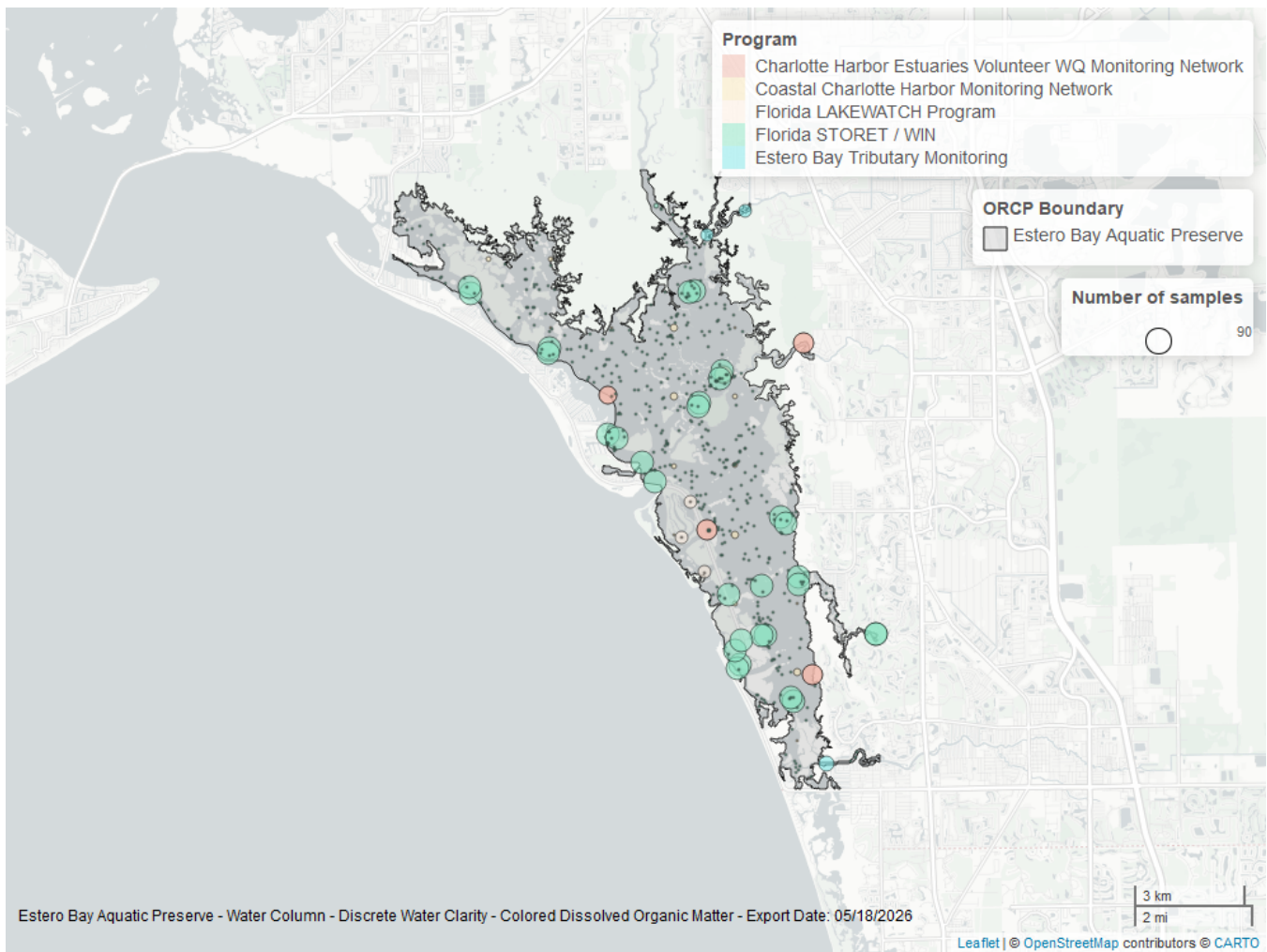


Figure 6: Map showing location of discrete water quality sampling locations within the boundaries of *Estero Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 11: Programs contributing data for Colored Dissolved Organic Matter

ProgramID	N_Data	YearMin	YearMax
5002	1768	2018	2025
476	301	2017	2025
4063	91	2018	2025
513	68	2003	2005
514	66	2011	2019

Program names:

5002 - Florida STORET / WIN¹

476 - Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network²

4063 - Estero Bay Tributary Monitoring³

513 - Coastal Charlotte Harbor Monitoring Network⁴

514 - Florida LAKEWATCH Program⁶

Dissolved Oxygen - Discrete

Seasonal Kendall-Tau Trend Analysis

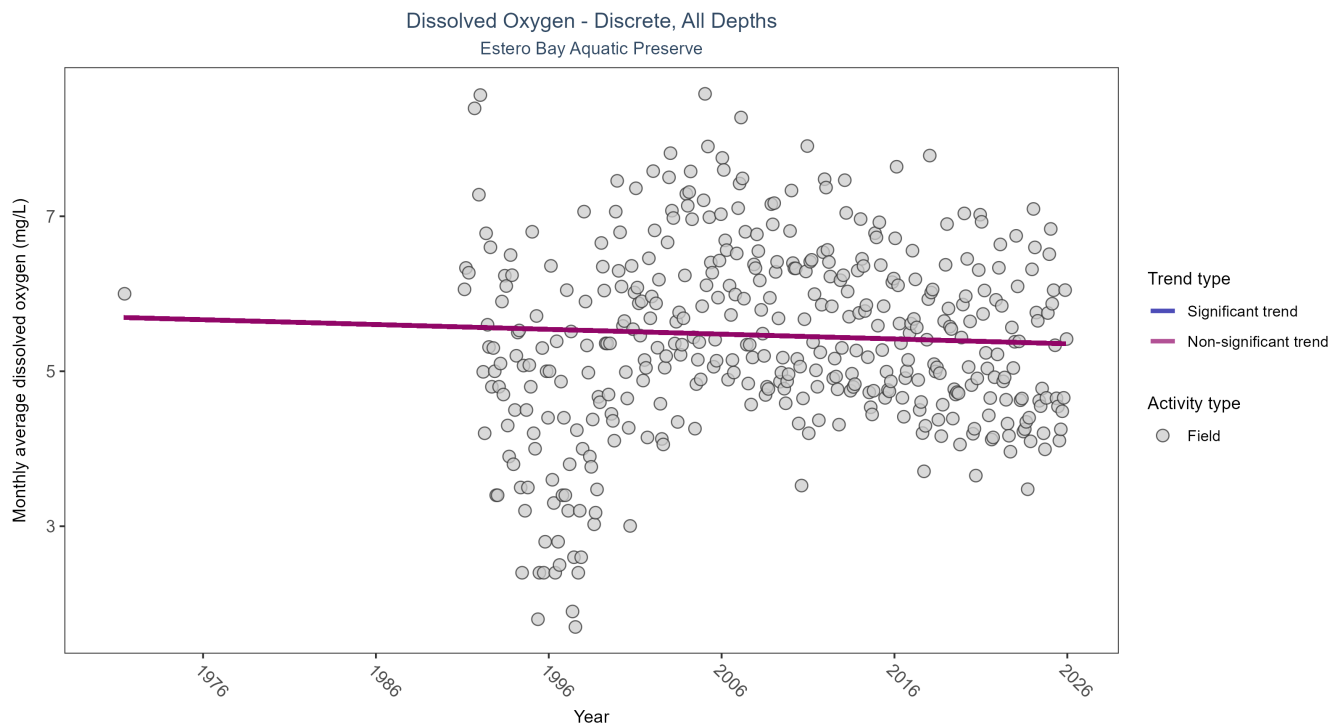


Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 12: Seasonal Kendall-Tau Trend Analysis for Dissolved Oxygen

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	11723	36	1971 - 2025	5.76	-0.0528	5.6962	-0.0062	0.1334

Dissolved oxygen showed no detectable trend between 1971 and 2025.

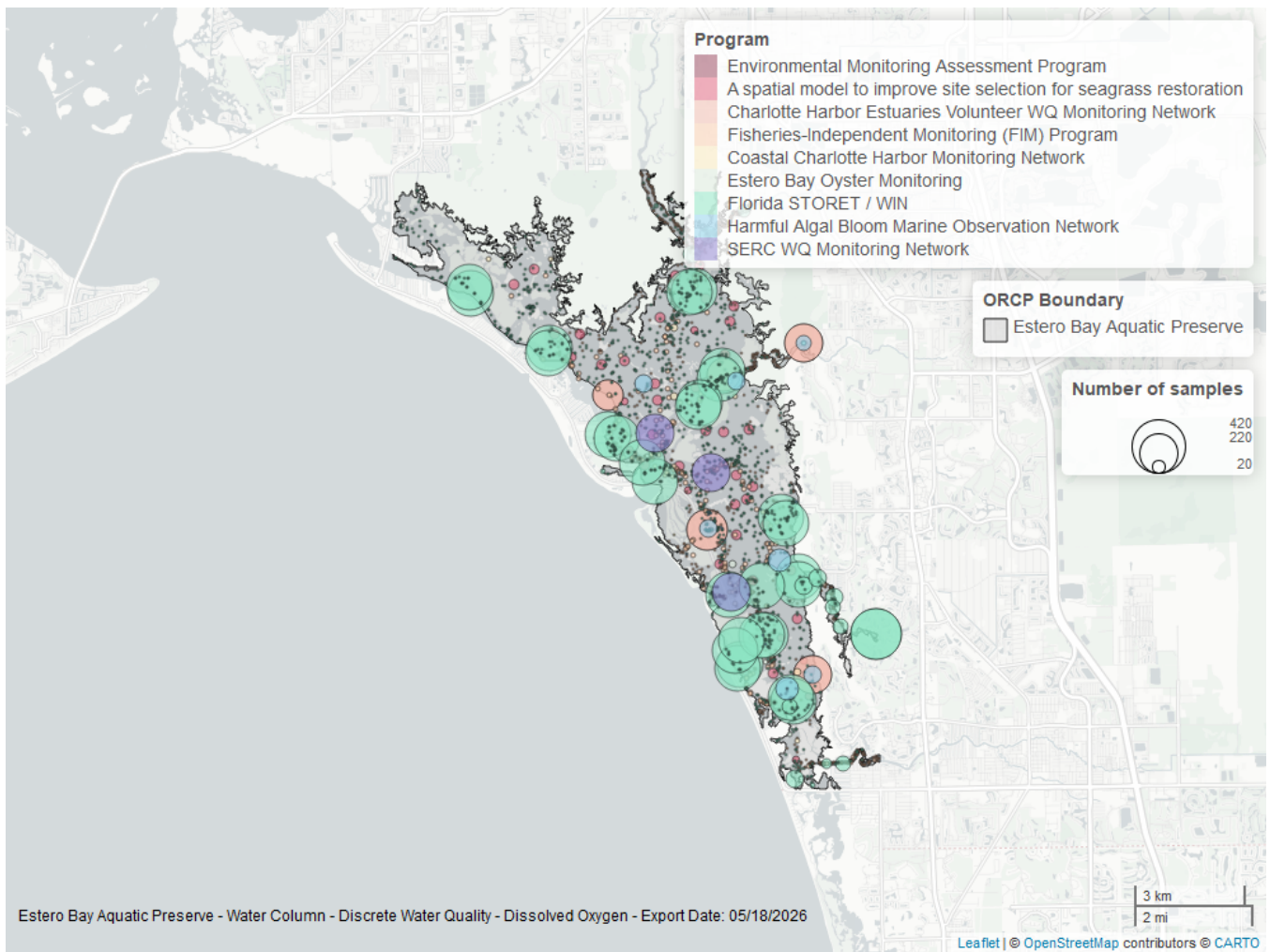


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Estero Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 13: Programs contributing data for Dissolved Oxygen

ProgramID	N_Data	YearMin	YearMax
5002	6744	1991	2025
69	2263	2001	2007
476	1035	1998	2025
509	696	1999	2008
4064	619	2011	2012
95	427	1971	2018
513	69	2003	2005
4042	62	2016	2024
115	3	2003	2003

Program names:

5002 - Florida STORET / WIN¹

69 - Fisheries-Independent Monitoring (FIM) Program⁹

476 - Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network²

509 - SERC Water Quality Monitoring Network⁵

4064 - A spatial model to improve site selection for seagrass restoration in shallow boating environments¹⁰
95 - Harmful Algal Bloom Marine Observation Network¹¹
513 - Coastal Charlotte Harbor Monitoring Network⁴
4042 - Estero Bay Oyster Monitoring¹²
115 - Environmental Monitoring Assessment Program⁷

Dissolved Oxygen Saturation - Discrete

Seasonal Kendall-Tau Trend Analysis

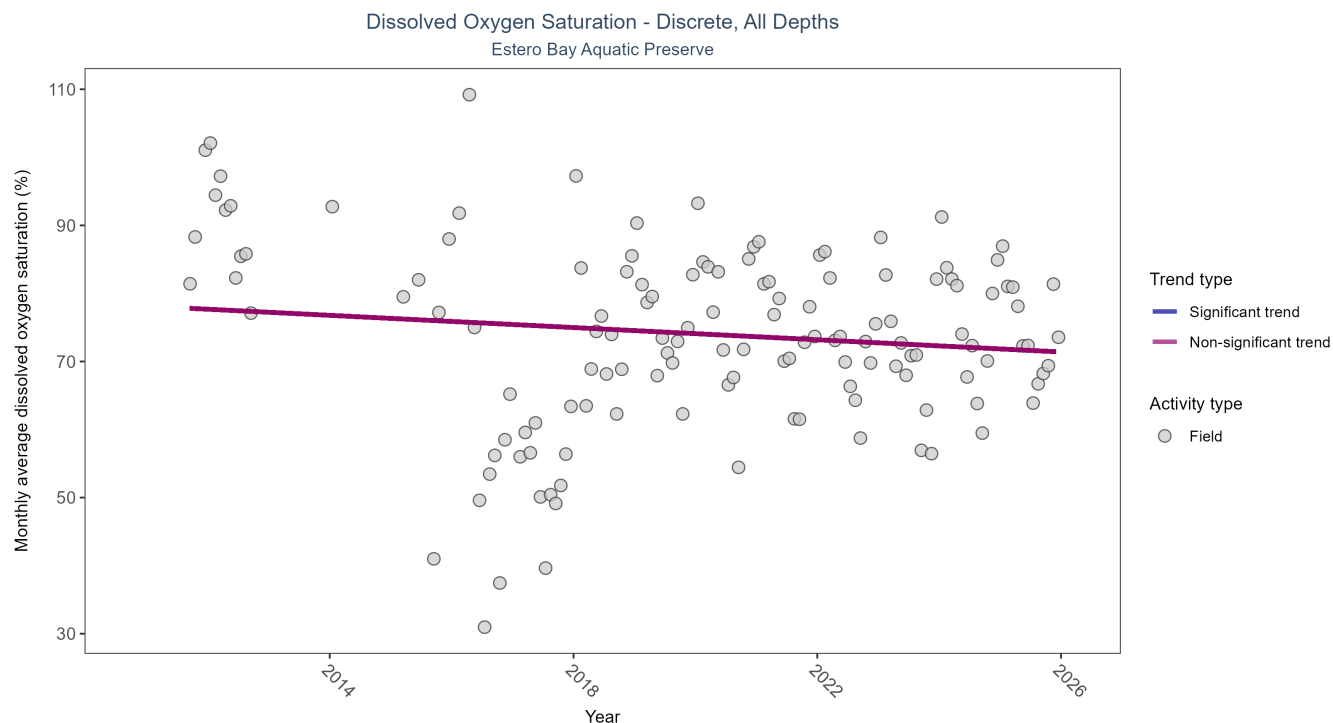


Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 14: Seasonal Kendall-Tau Trend Analysis for Dissolved Oxygen Saturation

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	3076	14	2011 - 2025	81.65	-0.1244	78.1287	-0.4489	0.0666

Dissolved oxygen saturation showed no detectable trend between 2011 and 2025.

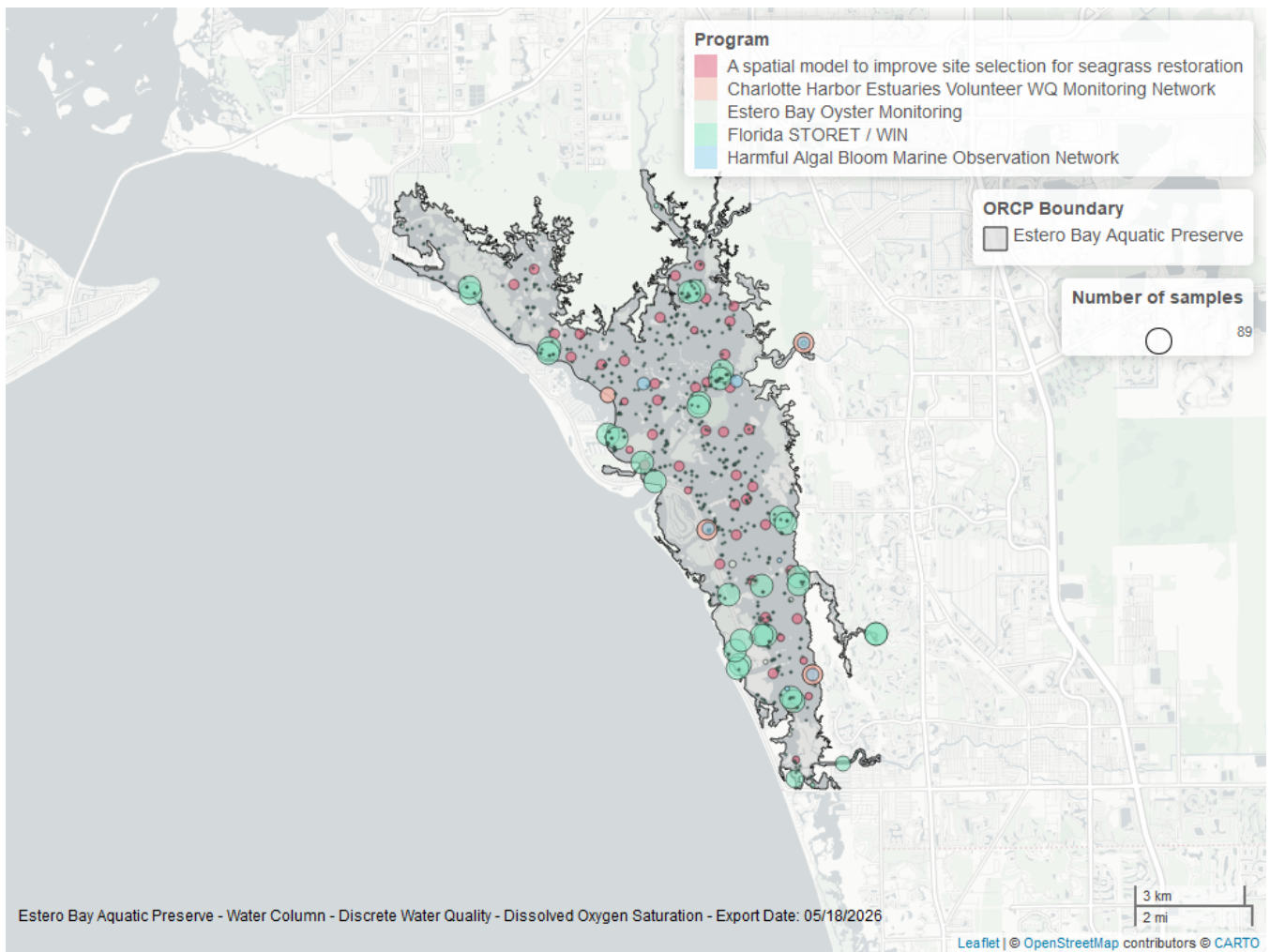


Figure 10: Map showing location of discrete water quality sampling locations within the boundaries of *Estero Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 15: Programs contributing data for Dissolved Oxygen Saturation

ProgramID	N_Data	YearMin	YearMax
5002	2036	2015	2025
4064	619	2011	2012
476	262	2017	2025
95	120	2011	2018
4042	53	2016	2024

Program names:

5002 - Florida STORET / WIN¹

4064 - A spatial model to improve site selection for seagrass restoration in shallow boating environments¹⁰

476 - Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network²

95 - Harmful Algal Bloom Marine Observation Network¹¹

4042 - Estero Bay Oyster Monitoring¹²

pH - Discrete

Seasonal Kendall-Tau Trend Analysis

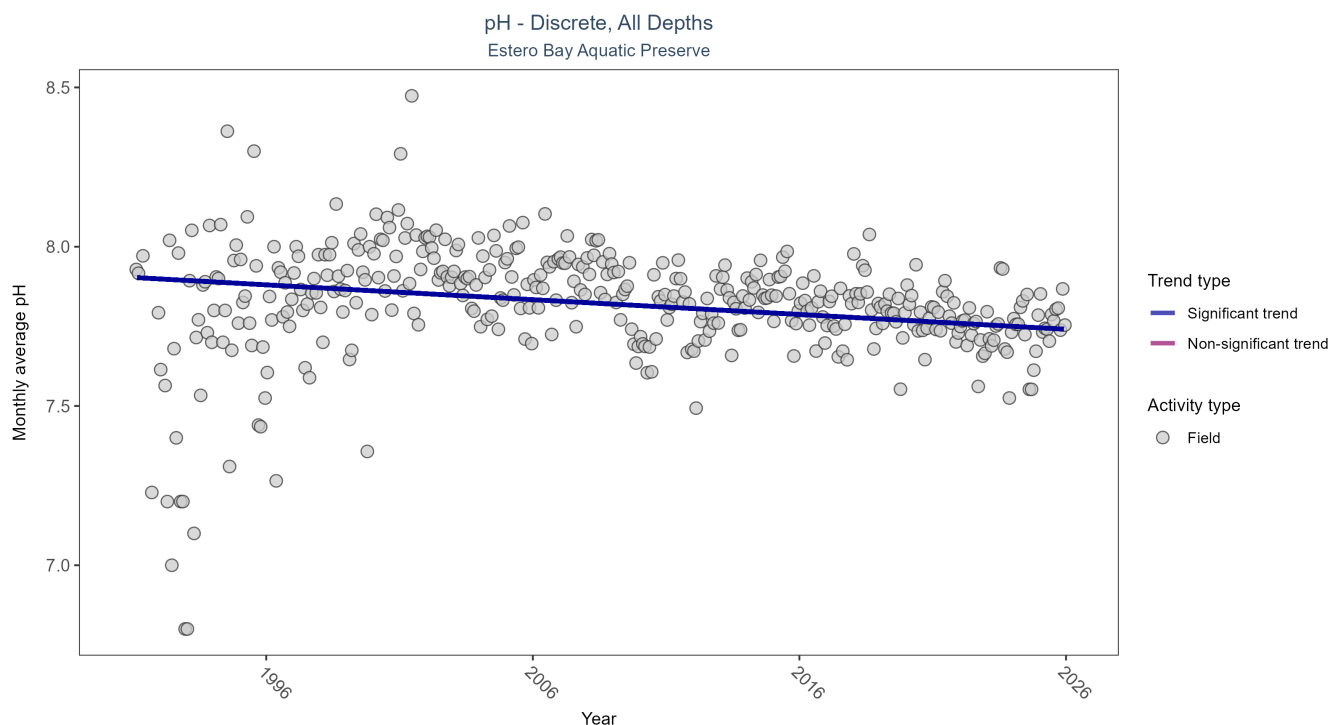


Figure 11: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 16: Seasonal Kendall-Tau Trend Analysis for pH

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	11094	35	1991 - 2025	7.9	-0.2201	7.9035	-0.0046	0

Monthly average pH decreased by less than 0.01 pH units per year.

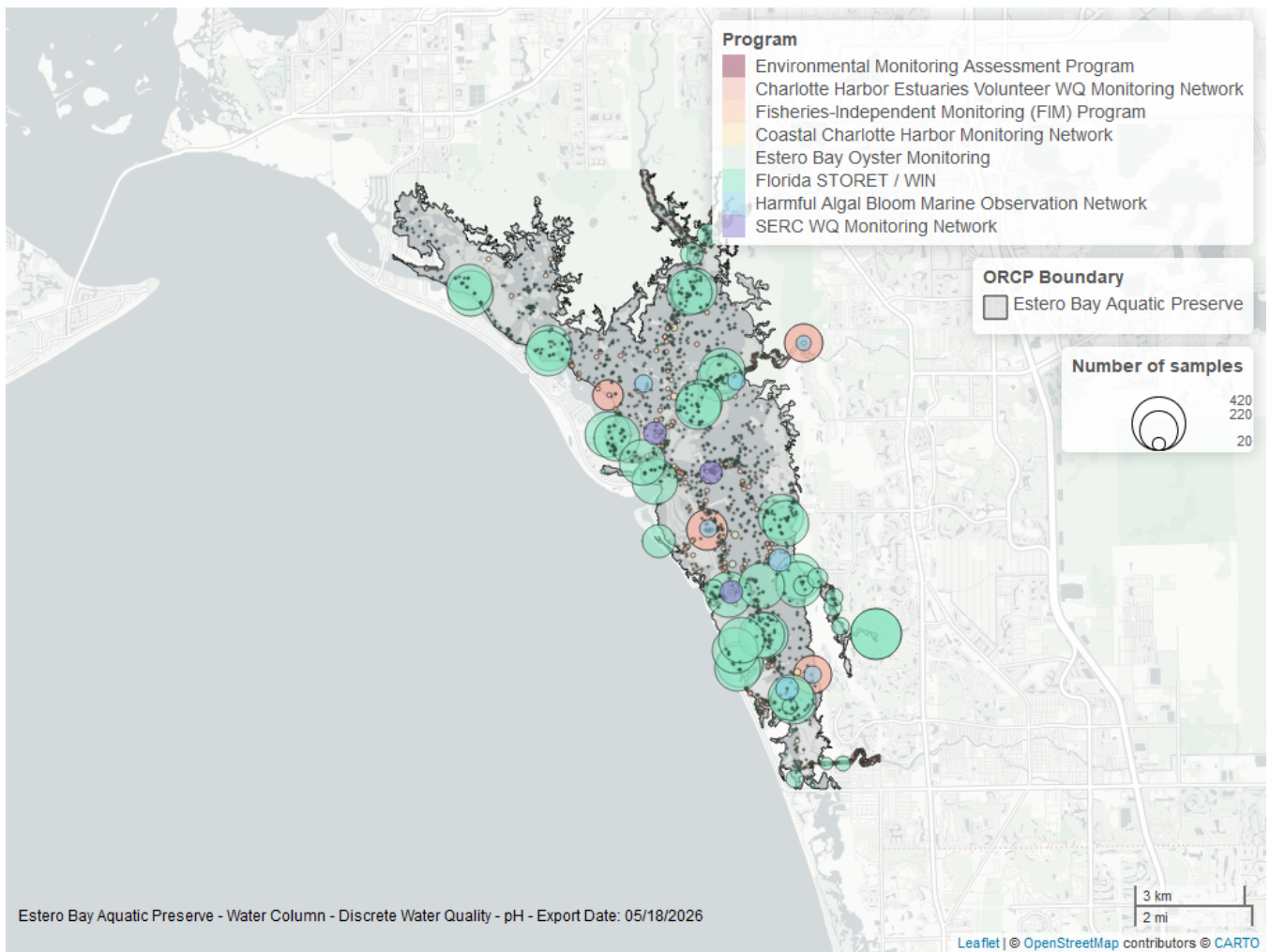


Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Estero Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 17: Programs contributing data for pH

ProgramID	N_Data	YearMin	YearMax
5002	7033	1991	2025
69	2264	2001	2007
476	1047	1998	2025
95	419	2005	2018
509	270	2001	2008
513	67	2003	2005
4042	56	2016	2024
115	3	2003	2003

Program names:

5002 - Florida STORET / WIN¹

69 - Fisheries-Independent Monitoring (FIM) Program⁹

476 - Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network²

95 - Harmful Algal Bloom Marine Observation Network¹¹

509 - SERC Water Quality Monitoring Network⁵

513 - Coastal Charlotte Harbor Monitoring Network⁴
4042 - Estero Bay Oyster Monitoring¹²
115 - Environmental Monitoring Assessment Program⁷

Salinity - Discrete

Seasonal Kendall-Tau Trend Analysis

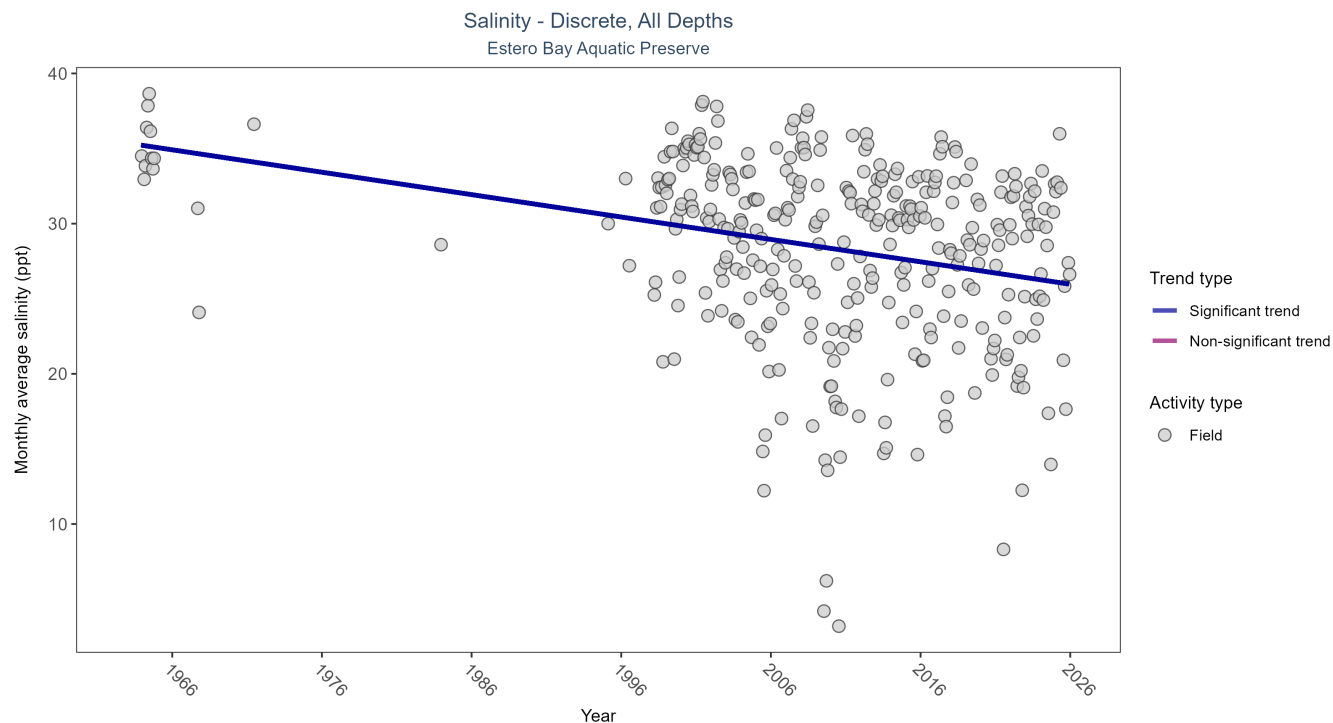


Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 18: Seasonal Kendall-Tau Trend Analysis for Salinity

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	Decreasing trend	5375	35	1963 - 2025	32.2	-0.2694	35.3624	-0.1492	0

Monthly average salinity decreased by 0.15 ppt per year.

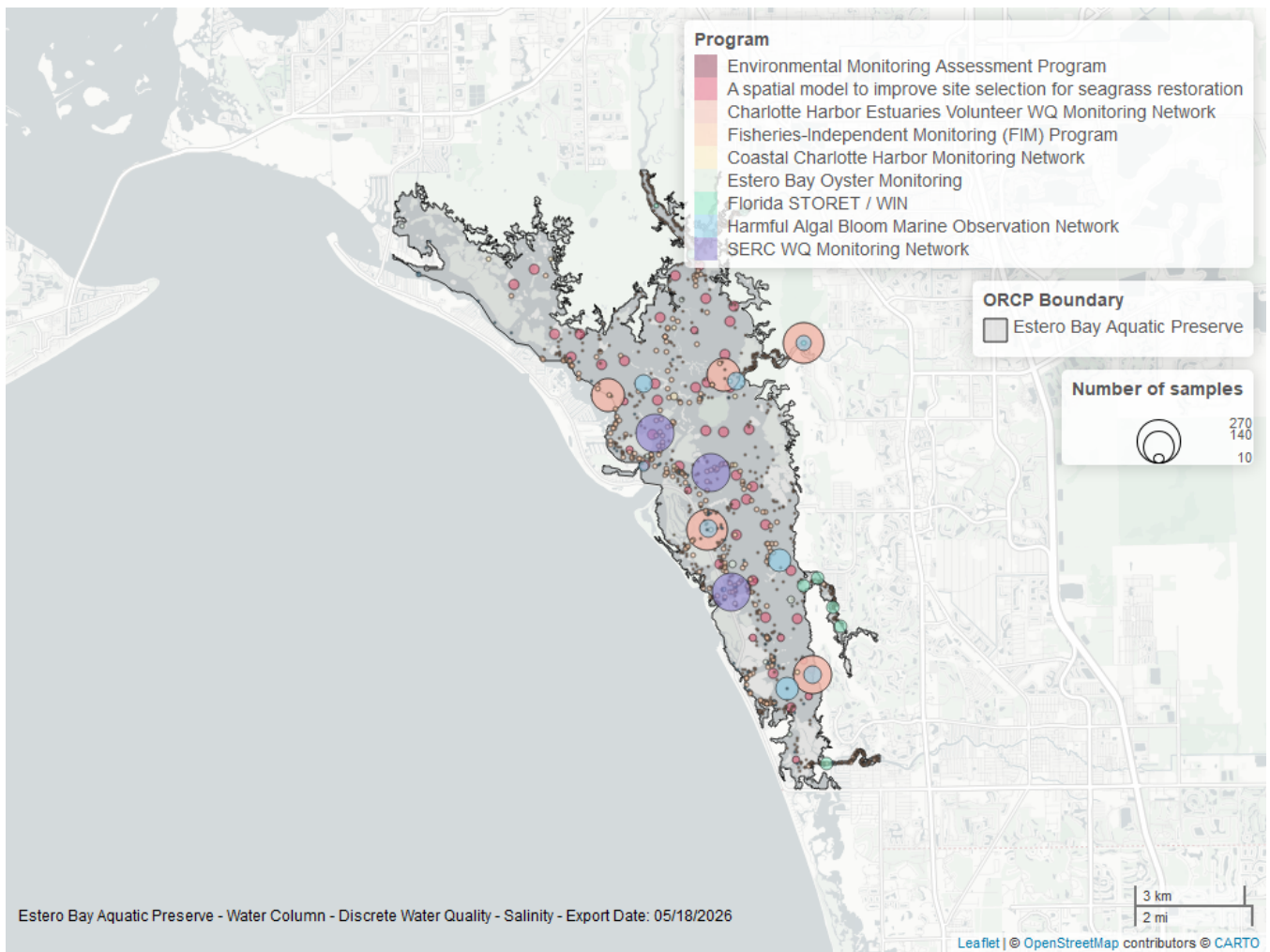


Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Estero Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 19: Programs contributing data for Salinity

ProgramID	N_Data	YearMin	YearMax
69	2263	2001	2007
476	1062	1998	2025
509	702	1999	2008
4064	619	2011	2012
95	501	1963	2018
5002	114	2009	2023
4042	62	2016	2024
513	60	2003	2005
115	3	2003	2003

Program names:

69 - Fisheries-Independent Monitoring (FIM) Program⁹

476 - Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network²

509 - SERC Water Quality Monitoring Network⁵

4064 - A spatial model to improve site selection for seagrass restoration in shallow boating environments¹⁰

95 - Harmful Algal Bloom Marine Observation Network¹¹
5002 - Florida STORET / WIN¹
~~4042~~ - Estero Bay Oyster Monitoring¹²
513 - Coastal Charlotte Harbor Monitoring Network⁴
115 - Environmental Monitoring Assessment Program⁷

Total Nitrogen - Discrete

Total Nitrogen Calculation:

The logic for calculated Total Nitrogen was provided by Kevin O'Donnell and colleagues at FDEP (with the help of Jay Silvanima, Watershed Monitoring Section). The following logic is used, in this order, based on the availability of specific nitrogen components.

- 1) $TN = TKN + NO3O2$;
- 2) $TN = TKN + NO3 + NO2$;
- 3) $TN = ORGN + NH4 + NO3O2$;
- 4) $TN = ORGN + NH4 + NO2 + NO3$;
- 5) $TN = TKN + NO3$;
- 6) $TN = ORGN + NH4 + NO3$;

Additional Information:

- Rules for use of sample fraction:
 - Florida Department of Environmental Protection (FDEP) report that if both “Total” and “Dissolved” components are reported, only “Total” is used. If the total is not reported, then the dissolved components are used as a best available replacement.
 - Total nitrogen calculations are done using nitrogen components with the same sample fraction, nitrogen components with mixed total/dissolved sample fractions are not used. In other words, total nitrogen can be calculated when TKN and NO3O2 are both total sample fractions, or when both are dissolved sample fractions. *Future calculations of total nitrogen values may be based on components with mixed sample fractions.*
- Values inserted into data:
 - ParameterName = “Total Nitrogen”
 - SEACAR_QAQCFlagCode = “1Q”
 - SEACAR_QAQC_Description = “SEACAR Calculated”

Seasonal Kendall-Tau Trend Analysis

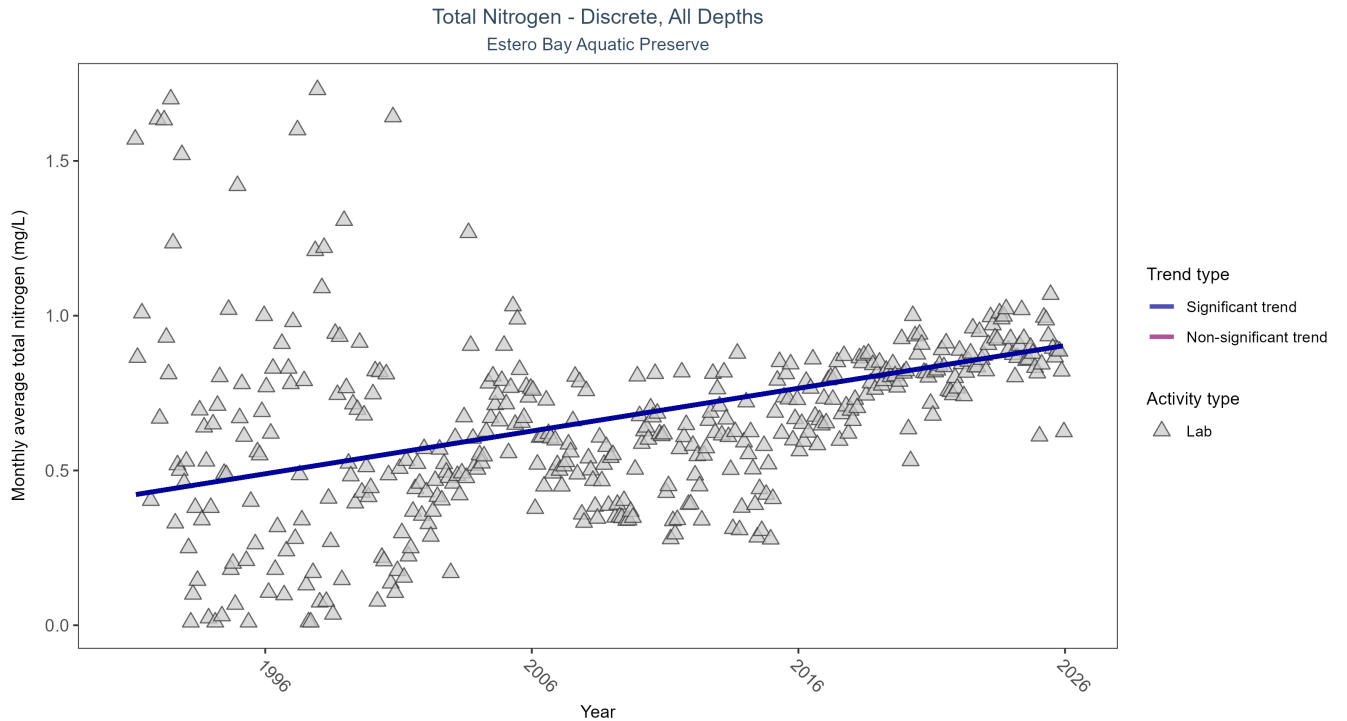


Figure 15: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 20: Seasonal Kendall-Tau Trend Analysis for Total Nitrogen

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	8274	35	1991 - 2025	0.6535	0.3433	0.4203	0.0138	0

Monthly average total nitrogen increased by 0.01 mg/L per year.

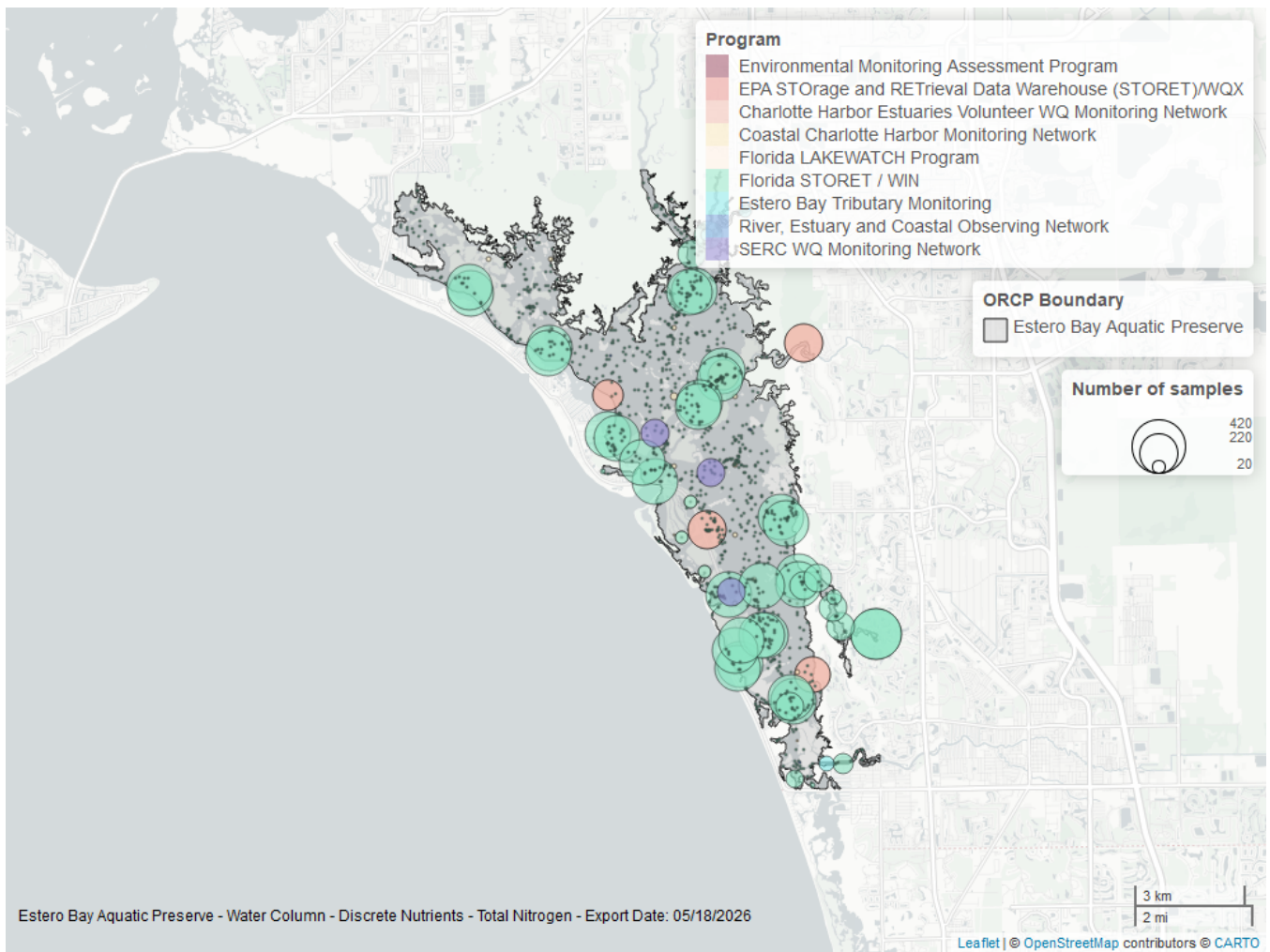


Figure 16: Map showing location of discrete water quality sampling locations within the boundaries of *Estero Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 21: Programs contributing data for Total Nitrogen

ProgramID	N_Data	YearMin	YearMax
5002	6759	1991	2025
476	958	1998	2025
509	351	1999	2008
514	84	2011	2019
4063	82	2018	2025
513	59	2003	2005
303	8	2020	2021
103	4	2003	2003
115	1	2003	2003

Program names:

5002 - Florida STORET / WIN¹

476 - Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network²

509 - SERC Water Quality Monitoring Network⁵

514 - Florida LAKEWATCH Program⁶

4063 - Estero Bay Tributary Monitoring³
 513 - Coastal Charlotte Harbor Monitoring Network⁴
 303 - River, Estuary and Coastal Observing Network¹³
 103 - EPA STORage and RETrieval Data Warehouse (STORET)/WQX⁸
 115 - Environmental Monitoring Assessment Program⁷

Total Phosphorus - Discrete

Seasonal Kendall-Tau Trend Analysis

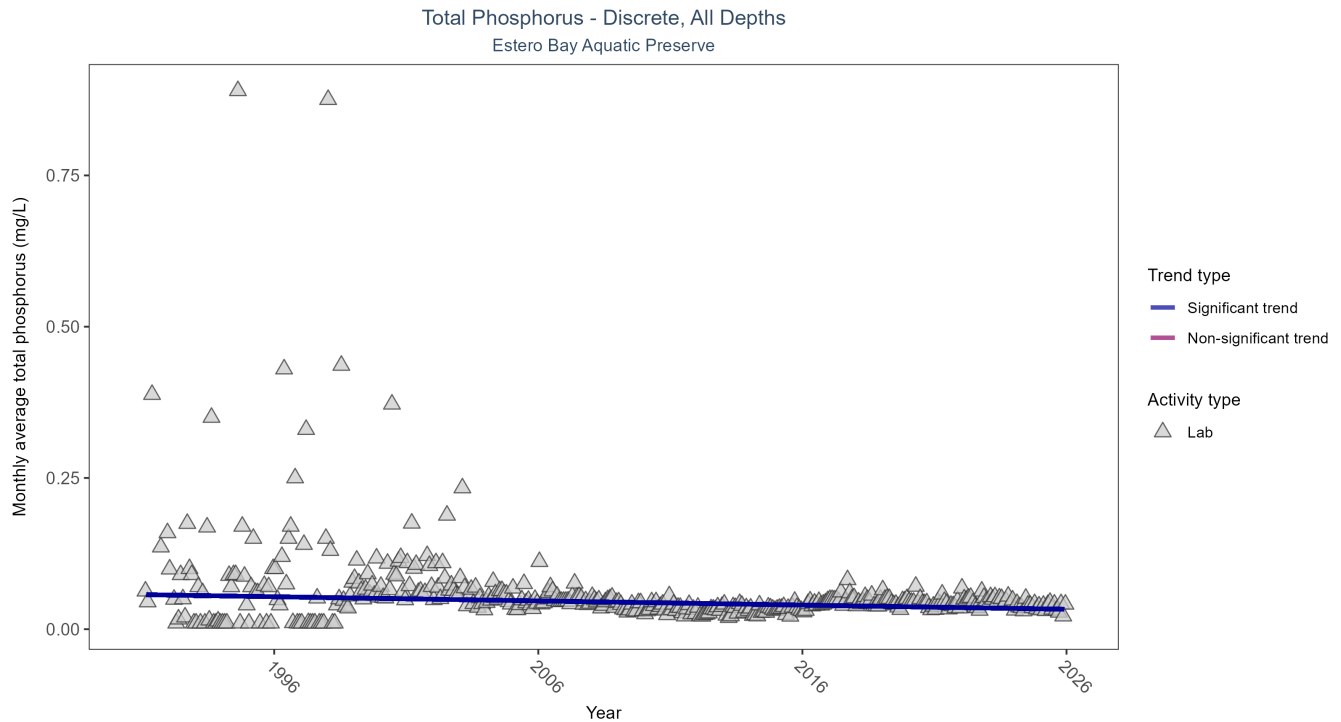


Figure 17: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 22: Seasonal Kendall-Tau Trend Analysis for Total Phosphorus

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	8279	35	1991 - 2025	0.041	-0.2034	0.0571	-0.0007	0

Monthly average total phosphorus decreased by less than 0.01 mg/L per year.

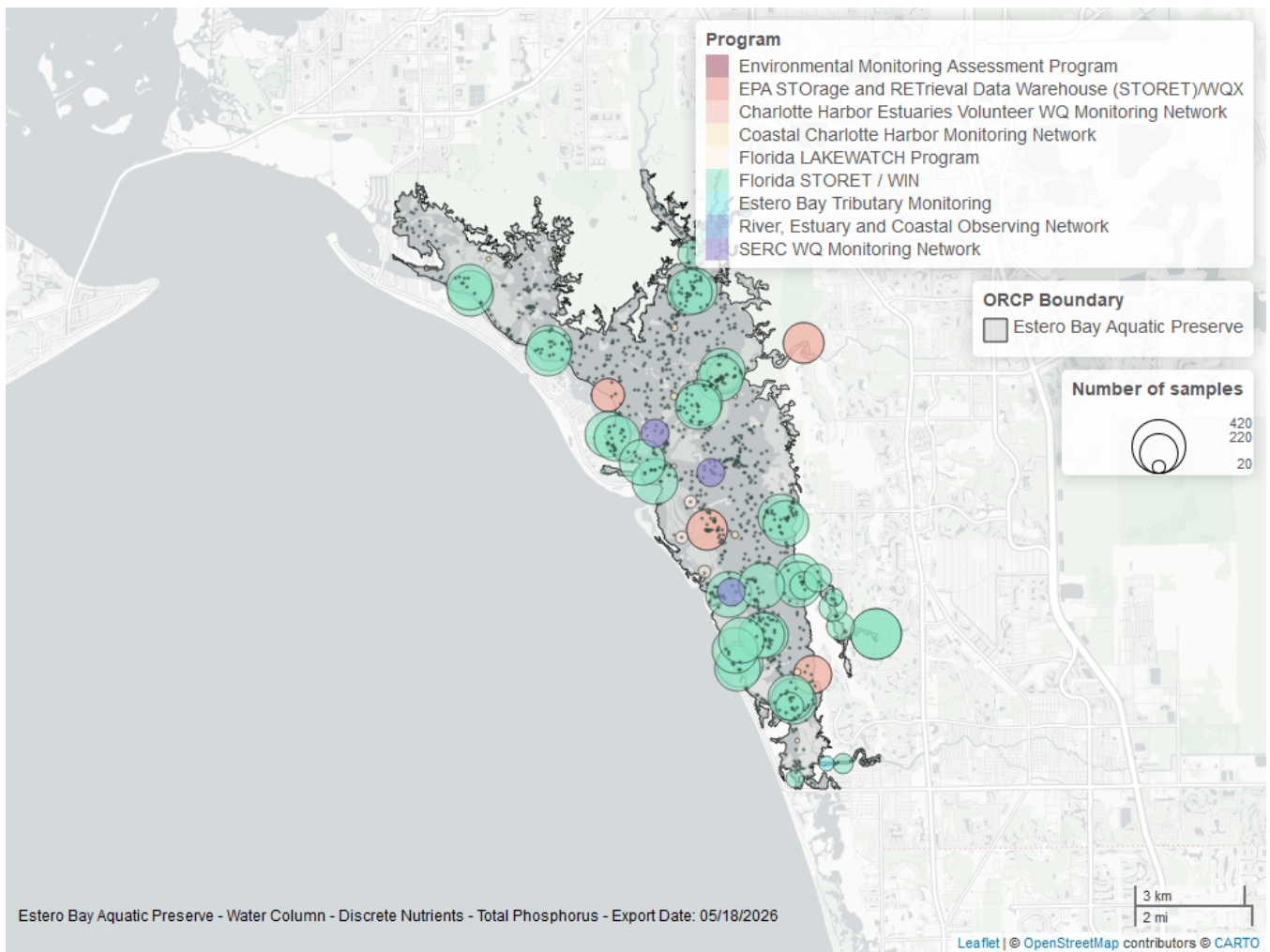


Figure 18: Map showing location of discrete water quality sampling locations within the boundaries of *Estero Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 23: Programs contributing data for Total Phosphorus

ProgramID	N_Data	YearMin	YearMax
5002	6688	1991	2025
476	1055	1998	2025
509	351	1999	2008
4063	91	2018	2025
514	84	2011	2019
513	69	2003	2005
303	8	2020	2021
103	3	2003	2003
115	1	2003	2003

Program names:

5002 - Florida STORET / WIN¹

476 - Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network²

509 - SERC Water Quality Monitoring Network⁵

4063 - Estero Bay Tributary Monitoring³

514 - Florida LAKEWATCH Program⁶
 513 - Coastal Charlotte Harbor Monitoring Network⁴
 303 - River, Estuary and Coastal Observing Network¹³
 103 - EPA STORage and RETrieval Data Warehouse (STORET)/WQX⁸
 115 - Environmental Monitoring Assessment Program⁷

Turbidity - Discrete

Seasonal Kendall-Tau Trend Analysis

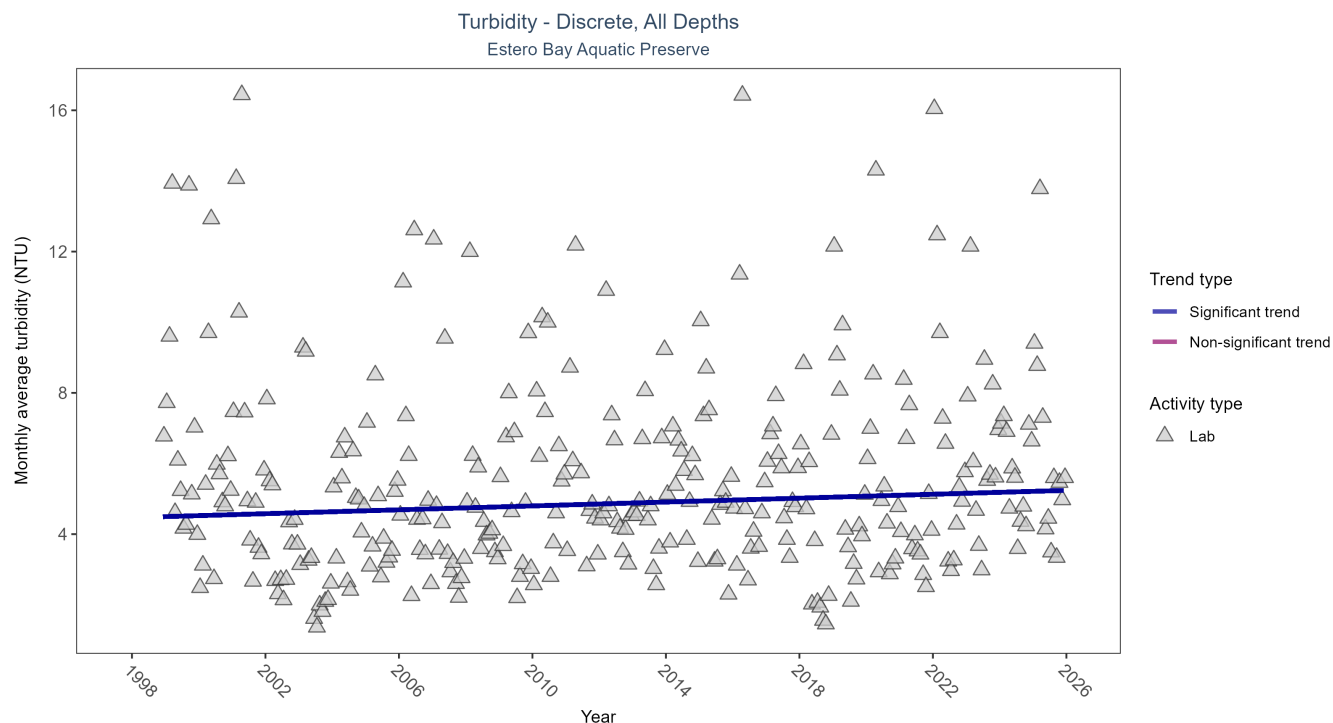


Figure 19: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 24: Seasonal Kendall-Tau Trend Analysis for Turbidity

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	3340	28	1998 - 2025	4.09	0.0788	4.4687	0.0275	0.0414

Monthly average turbidity increased by 0.03 NTU per year, indicating a decrease in water clarity.

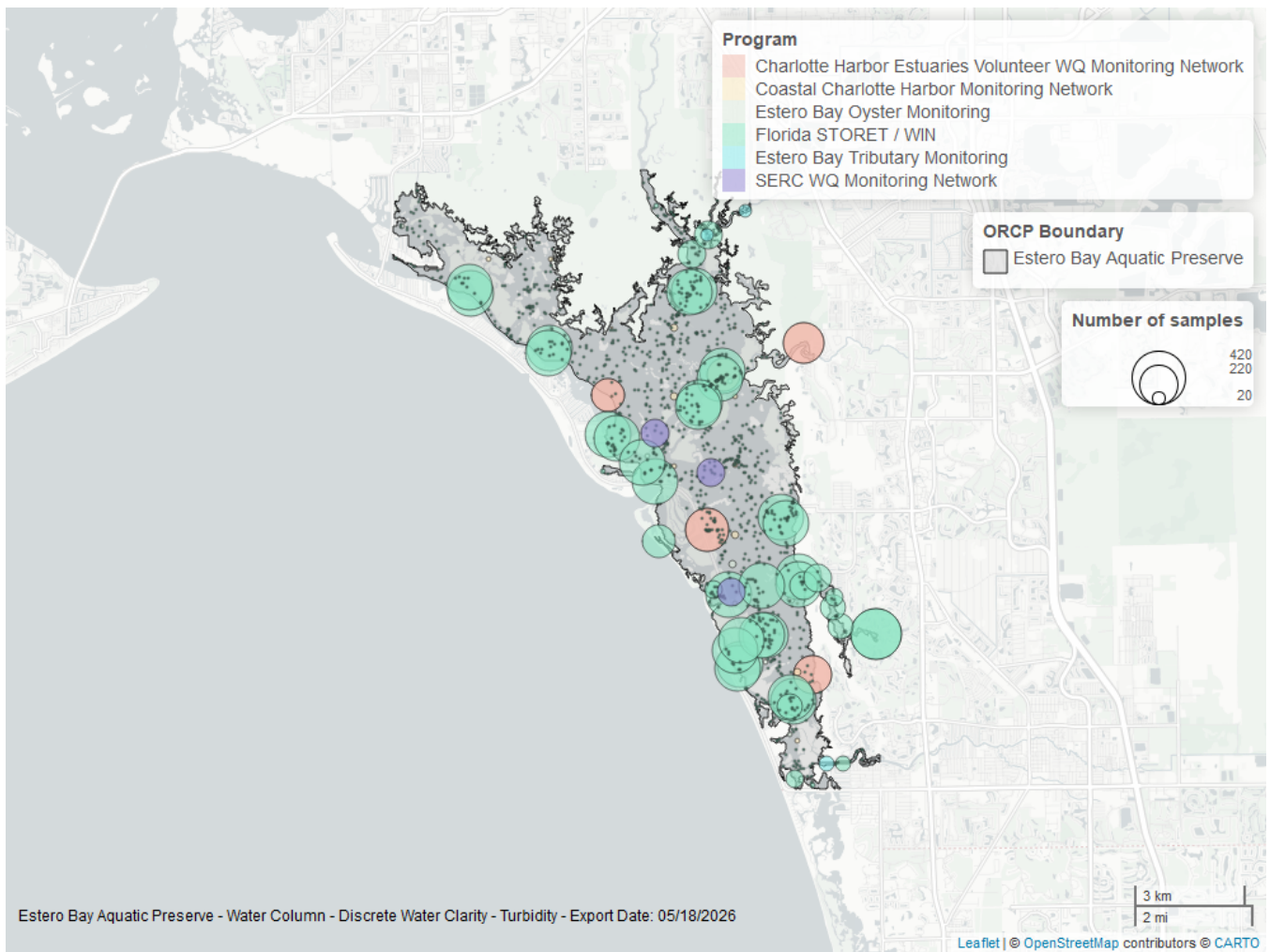


Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *Estero Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 25: Programs contributing data for Turbidity

ProgramID	N_Data	YearMin	YearMax
5002	6781	1991	2025
476	1111	1998	2025
509	348	1999	2008
4063	91	2018	2025
513	69	2003	2005
4042	61	2016	2024

Program names:

5002 - Florida STORET / WIN¹

476 - Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network²

509 - SERC Water Quality Monitoring Network⁵

4063 - Estero Bay Tributary Monitoring³

513 - Coastal Charlotte Harbor Monitoring Network⁴

4042 - Estero Bay Oyster Monitoring¹²

Water Temperature - Discrete

Seasonal Kendall-Tau Trend Analysis

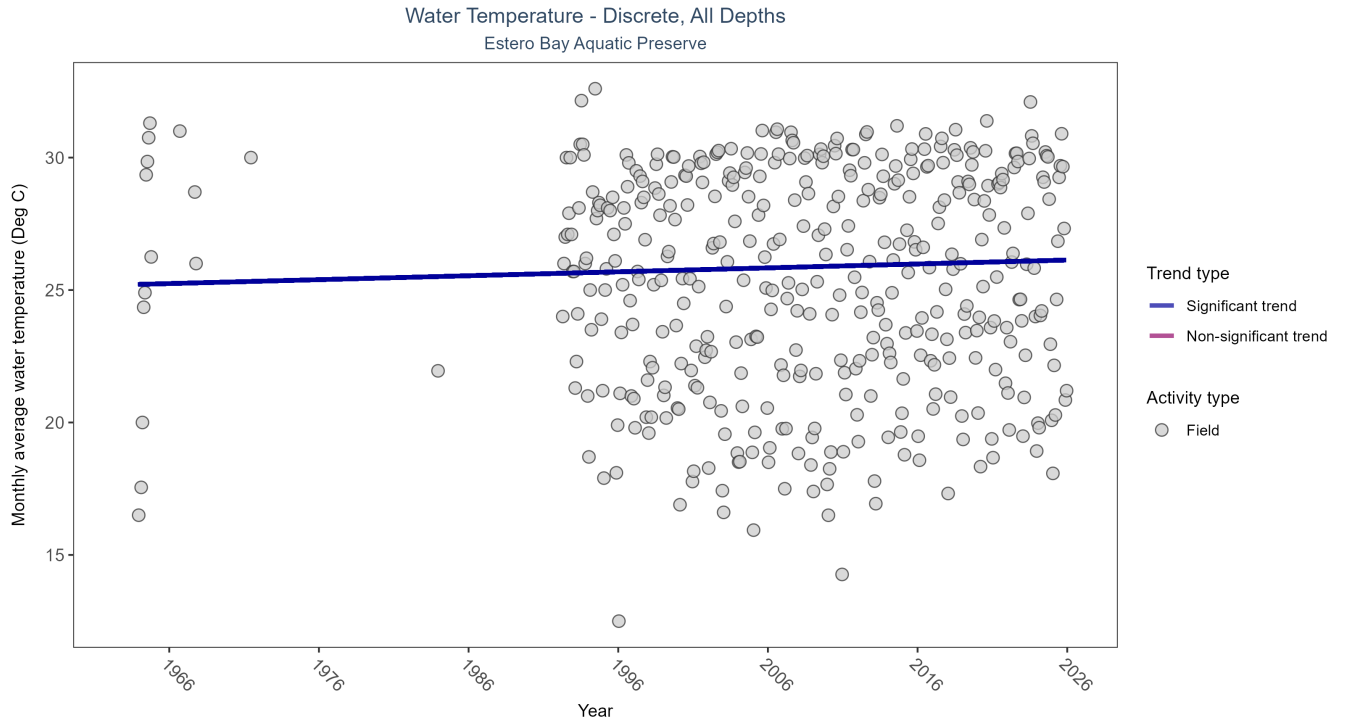


Figure 21: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 26: Seasonal Kendall-Tau Trend Analysis for Water Temperature

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	11361	40	1963 - 2025	26	0.0843	25.199	0.0148	0.0136

Monthly average water temperature increased by 0.01°C per year.

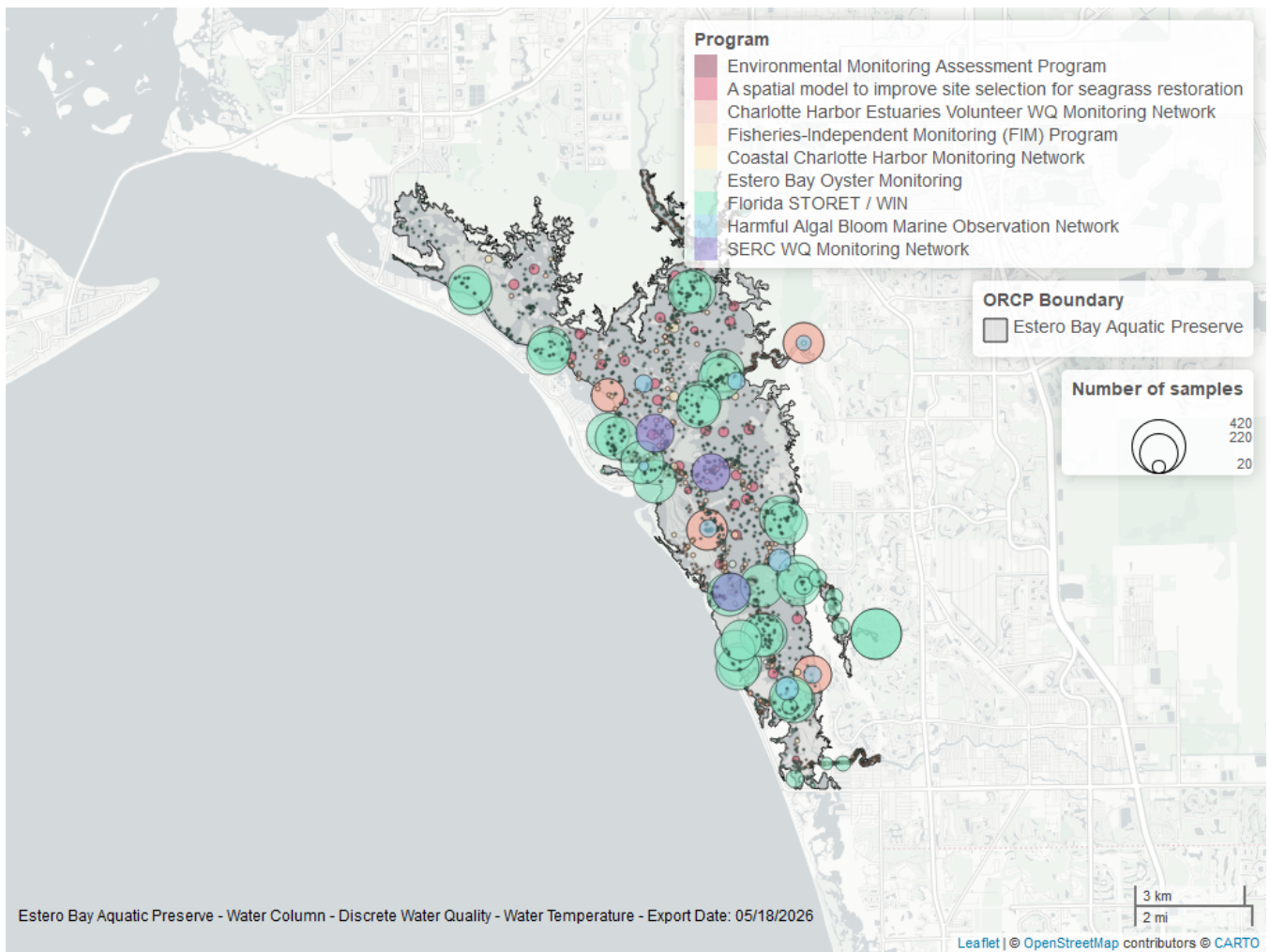


Figure 22: Map showing location of discrete water quality sampling locations within the boundaries of *Estero Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 27: Programs contributing data for Water Temperature

ProgramID	N_Data	YearMin	YearMax
5002	6052	1992	2025
69	2264	2001	2007
476	1068	1998	2025
509	702	1999	2008
4064	619	2011	2012
95	466	1963	2018
513	130	2003	2005
4042	62	2016	2024
115	3	2003	2003

Program names:

5002 - Florida STORET / WIN¹

69 - Fisheries-Independent Monitoring (FIM) Program⁹

476 - Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network²

509 - SERC Water Quality Monitoring Network⁵

~~4064~~ - A spatial model to improve site selection for seagrass restoration in shallow boating environments¹⁰
95 - Harmful Algal Bloom Marine Observation Network¹¹
513 - Coastal Charlotte Harbor Monitoring Network⁴
~~4042~~ - Estero Bay Oyster Monitoring¹²
115 - Environmental Monitoring Assessment Program⁷

Water Quality - Continuous

The following files were used in the continuous analysis:

- *Combined_WQ_WC_NUT_cont_Chlorophyll_a_Uncorrected_for_Pheophytin-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Dissolved_Oxygen-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Dissolved_Oxygen_Saturation-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Fluorescent_Dissolved_Organic_Matter-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_pH-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Salinity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Specific_Conductivity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Turbidity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Water_Temperature-2026-Mar-06.txt*

Continuous monitoring locations in Estero Bay Aquatic Preserve

Table 28: Station overview for Continuous parameters by Program

<i>ProgramID</i>	<i>ProgramLocationID</i>	<i>Years of Data</i>	<i>Use in Analysis</i>	<i>Parameters</i>
474	EB01	19	TRUE	pH , Dissolved Oxygen Saturation, Dissolved Oxygen , Turbidity , Water Temperature , Salinity , Specific Conductivity
474	EB01b	2	FALSE	pH , Dissolved Oxygen Saturation, Dissolved Oxygen , Turbidity , Water Temperature , Salinity , Specific Conductivity
474	EB02	22	TRUE	pH , Dissolved Oxygen Saturation, Dissolved Oxygen , Turbidity , Water Temperature , Salinity , Specific Conductivity
474	EB03	22	TRUE	pH , Dissolved Oxygen Saturation, Dissolved Oxygen , Turbidity , Water Temperature , Salinity , Specific Conductivity
474	EB04	5	TRUE	pH , Dissolved Oxygen Saturation, Dissolved Oxygen , Turbidity , Water Temperature , Salinity , Specific Conductivity

Program names:

474 - Estero Bay Aquatic Preserve Continuous Water Quality Monitoring¹⁴

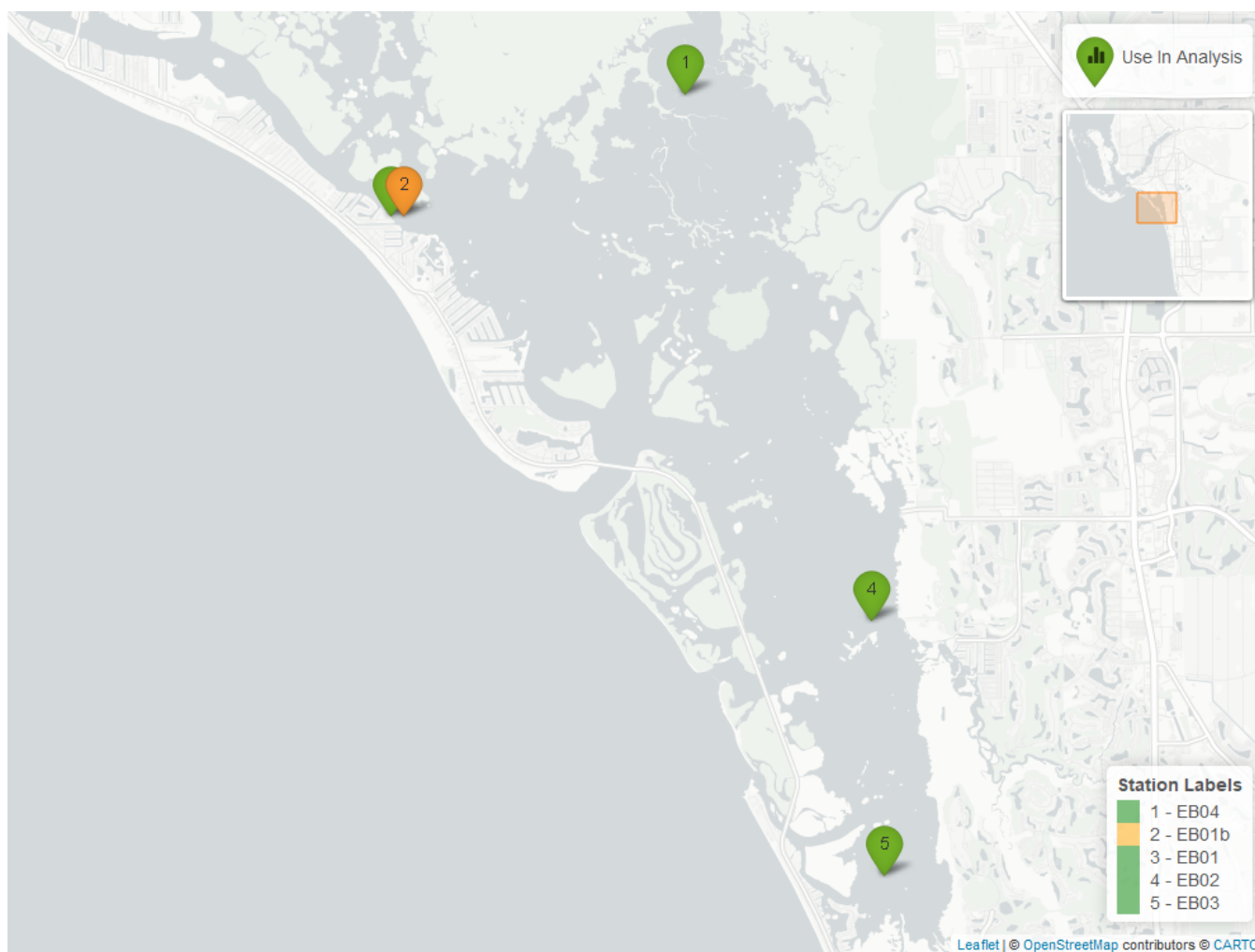


Figure 23: Map showing continuous water quality sampling locations within the boundaries of *Estero Bay Aquatic Preserve*. Sites marked as *Use In Analysis* (green) are featured in this report.

pH - Continuous

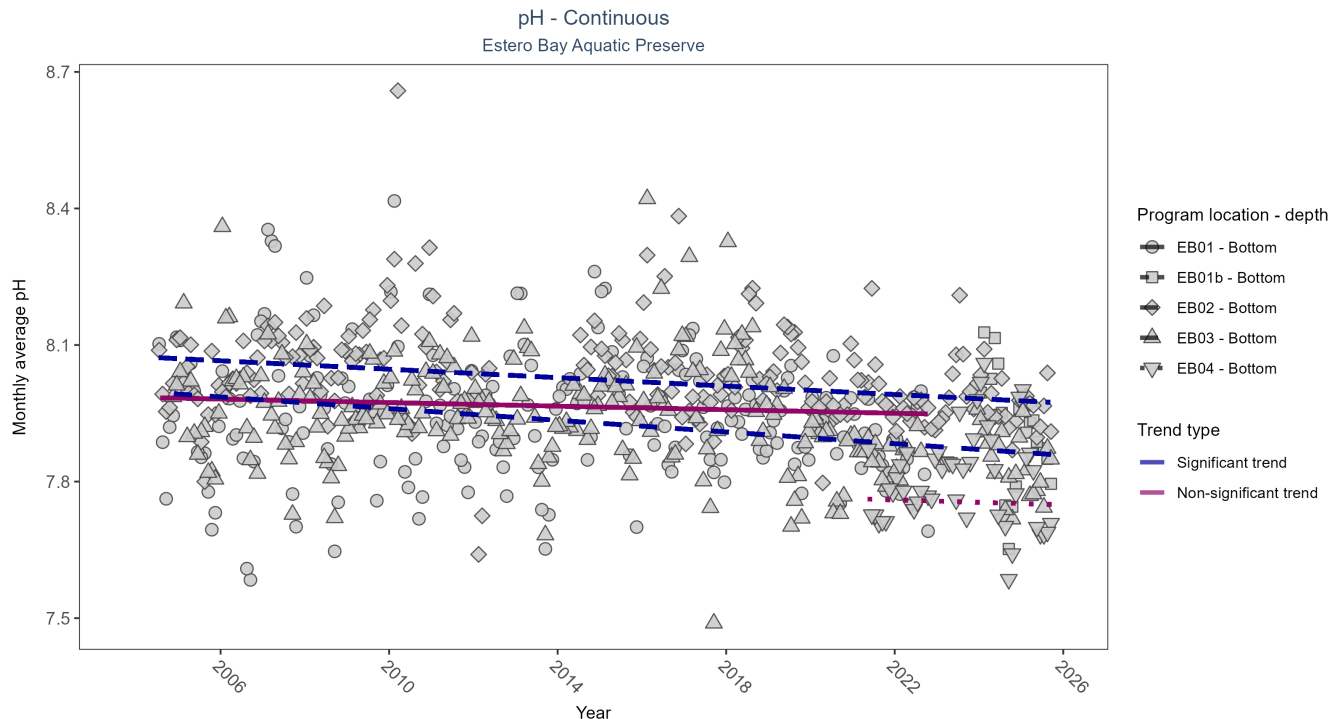


Figure 24: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 29: Seasonal Kendall-Tau Results for pH - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
EB01	No significant trend	562188	19	2004 - 2022	7.9	-0.08	7.99	0	0.12
EB01b	Insufficient data to calculate trend	55751	2	2024 - 2025	7.9	-	-	-	-
EB02	Significantly decreasing trend	580963	21	2004 - 2025	8.0	-0.24	8.07	0	0
EB03	Significantly decreasing trend	586236	22	2004 - 2025	7.9	-0.26	8	-0.01	0
EB04	No significant trend	148357	5	2021 - 2025	7.8	-0.02	7.76	0	0.67

At two program locations, monthly average pH decreased by less than 0.01 pH units per year at one site and by 0.01 pH units per year at the other. No detectable change in monthly average pH was observed at two locations. There was insufficient data to fit a model for one location.

Dissolved Oxygen Saturation - Continuous

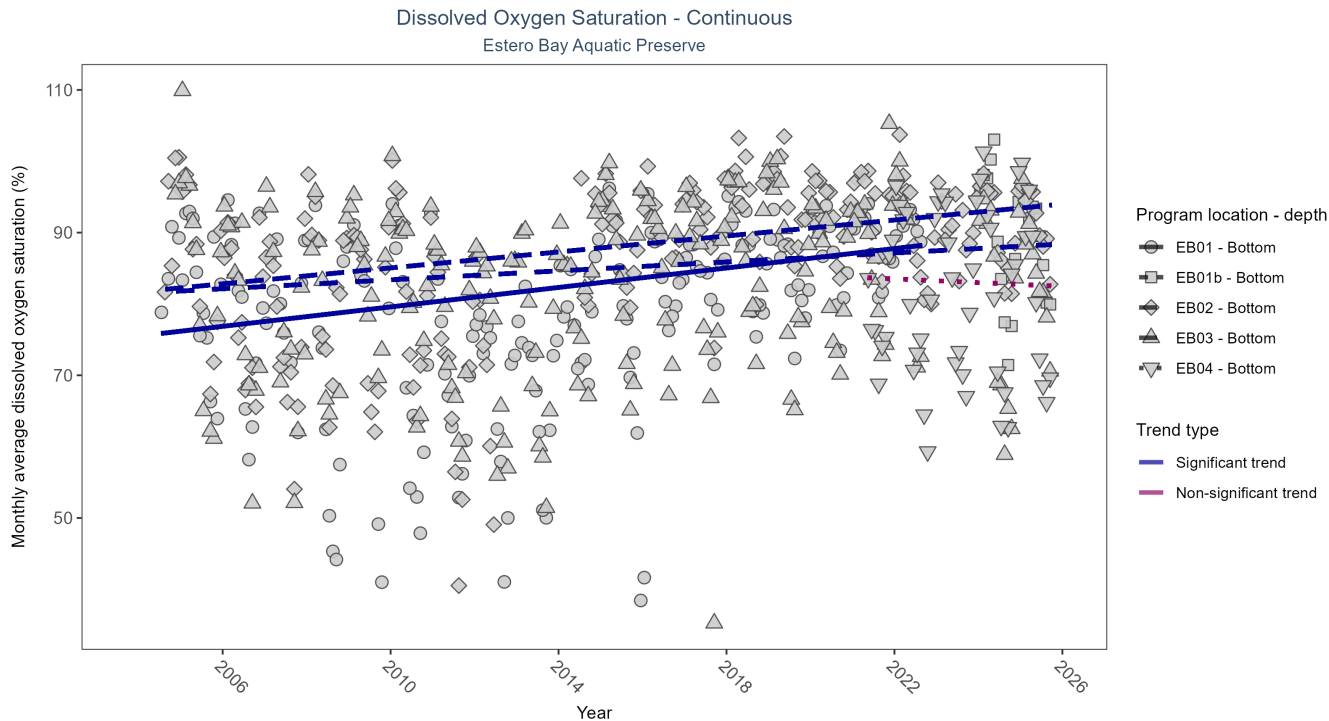


Figure 25: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 30: Seasonal Kendall-Tau Results for Dissolved Oxygen Saturation - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
EB01	Significantly increasing trend	480233	19	2004 - 2022	81.6	0.31	75.5	0.68	0
EB01b	Insufficient data to calculate trend	58532	2	2024 - 2025	88.9	-	-	-	-
EB02	Significantly increasing trend	506979	21	2004 - 2025	88.1	0.35	81.7	0.56	0
EB03	Significantly increasing trend	500104	21	2004 - 2025	83.7	0.21	81.5	0.31	0
EB04	No significant trend	143845	5	2021 - 2025	82.4	0.03	83.77	-0.26	0.93

At three program locations, monthly average dissolved oxygen saturation increased between 0.31 and 0.68% per year. No detectable change in monthly average dissolved oxygen saturation was observed at one location. There was insufficient data to fit a model for one location.

Dissolved Oxygen - Continuous

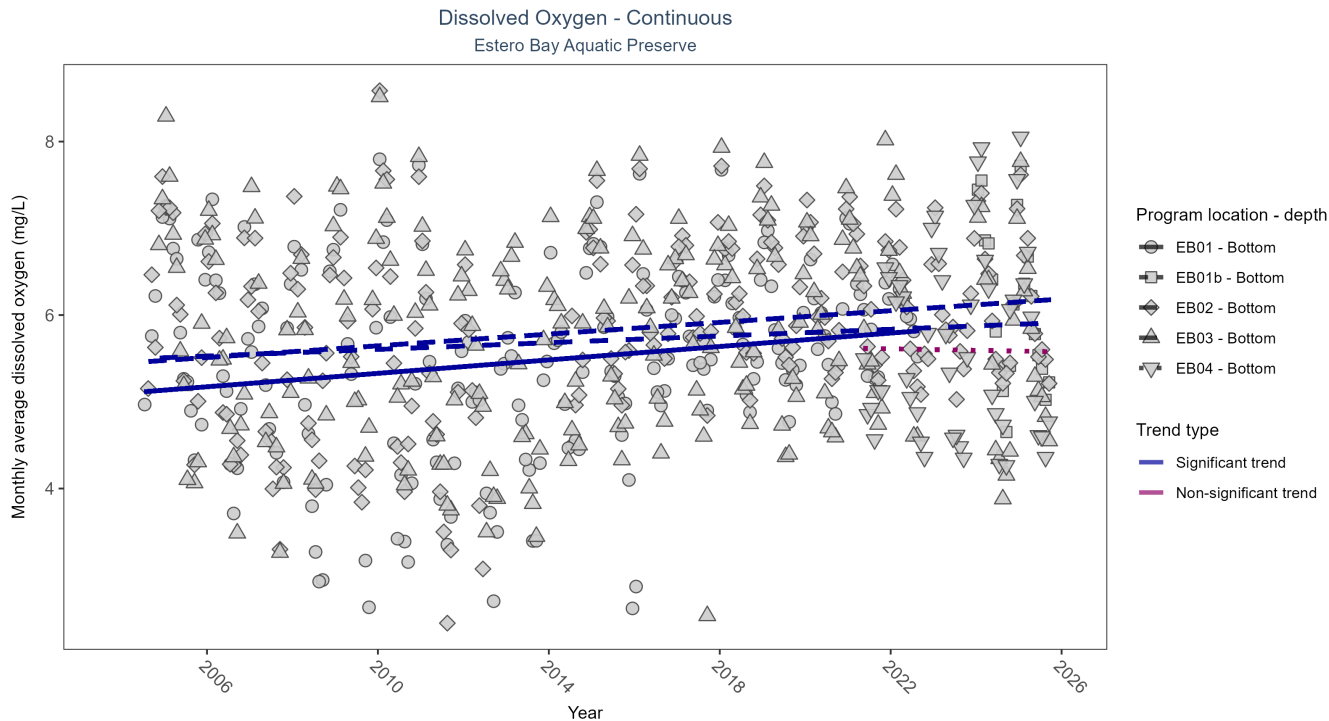


Figure 26: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 31: Seasonal Kendall-Tau Results for Dissolved Oxygen - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
EB01	Significantly increasing trend	478415	19	2004 - 2022	5.6	0.22	5.1	0.04	0
EB01b	Insufficient data to calculate trend	57822	2	2024 - 2025	6.1	-	-	-	-
EB02	Significantly increasing trend	506742	21	2004 - 2025	6.1	0.24	5.44	0.03	0
EB03	Significantly increasing trend	498578	21	2004 - 2025	5.9	0.15	5.49	0.02	0
EB04	No significant trend	132011	5	2021 - 2025	5.7	0.07	5.62	-0.01	1

At three program locations, monthly average dissolved oxygen increased between 0.02 and 0.04 mg/L per year. No detectable change in monthly average dissolved oxygen was observed at one location. There was insufficient data to fit a model for one location.

Turbidity - Continuous

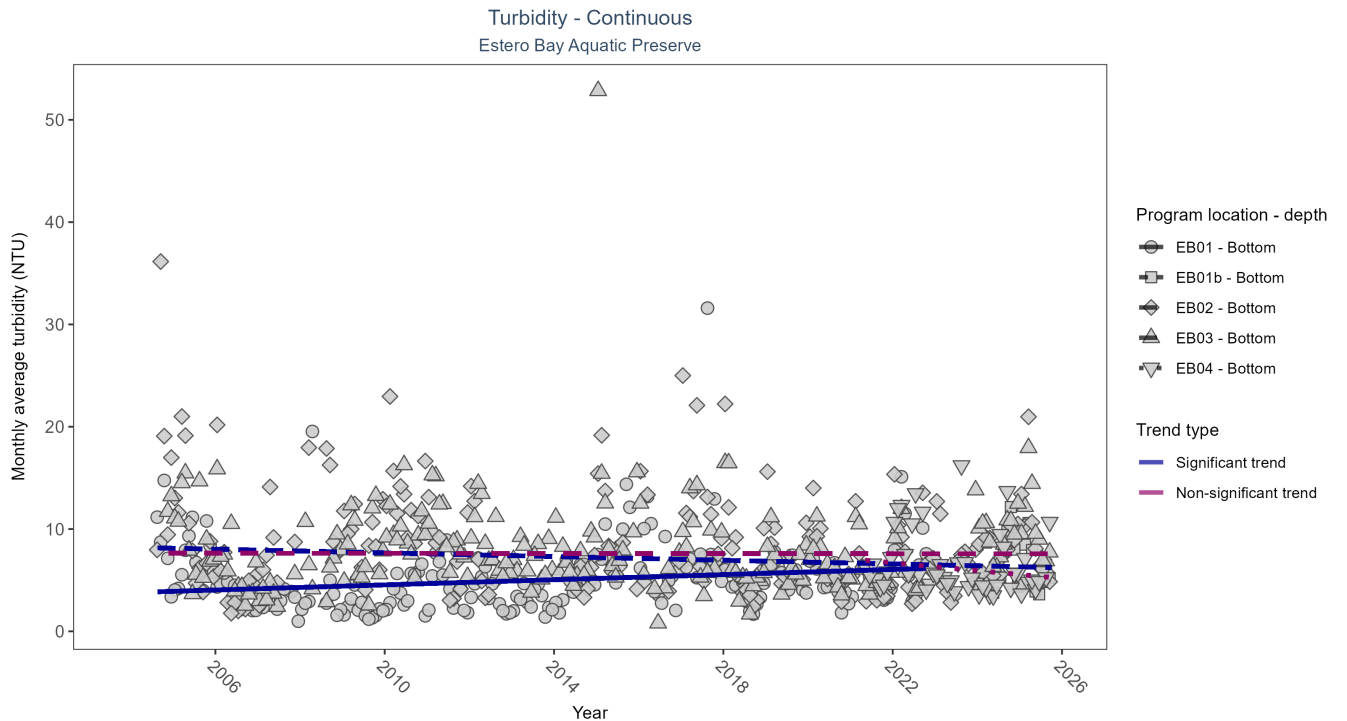


Figure 27: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 32: Seasonal Kendall-Tau Results for Turbidity - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
EB01	Significantly increasing trend	510965	19	2004 - 2022	4	0.18	3.79	0.13	0
EB01b	Insufficient data to calculate trend	55382	2	2024 - 2025	5	-	-	-	-
EB02	Significantly decreasing trend	493723	21	2004 - 2025	5	-0.12	8.21	-0.09	0.03
EB03	No significant trend	474845	22	2004 - 2025	6	-0.01	7.64	0	0.96
EB04	No significant trend	139581	5	2021 - 2025	5	-0.26	7.08	-0.38	0.08

At one program location, monthly average turbidity increased by 0.13 NTU per year. At one program location, monthly average turbidity decreased by 0.09 NTU per year. No detectable change in monthly average turbidity was observed at two locations. There was insufficient data to fit a model for one location.

Water Temperature - Continuous

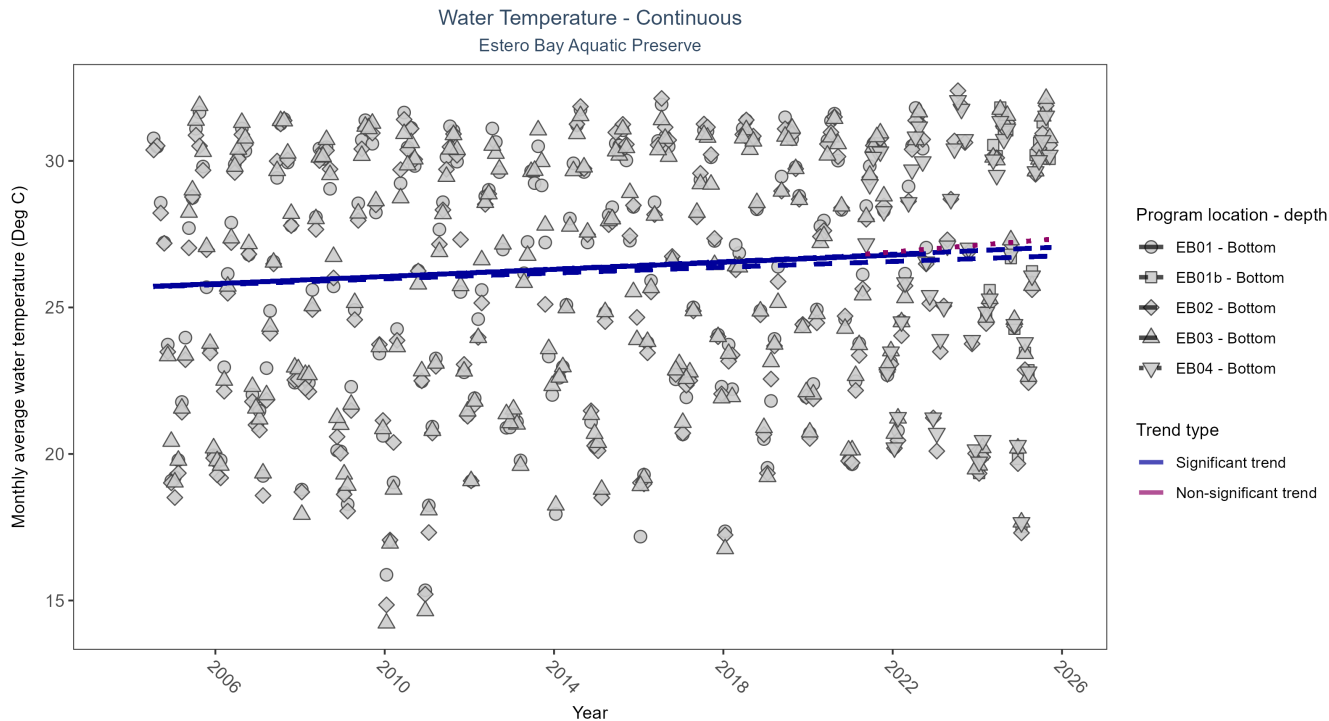


Figure 28: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 33: Seasonal Kendall-Tau Results for Water Temperature - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
EB01	Significantly increasing trend	617636	19	2004 - 2022	26.8	0.25	25.69	0.06	0
EB01b	Insufficient data to calculate trend	58530	2	2024 - 2025	28.0	-	-	-	-
EB02	Significantly increasing trend	643700	22	2004 - 2025	26.6	0.31	25.68	0.06	0
EB03	Significantly increasing trend	633752	22	2004 - 2025	26.5	0.2	25.68	0.05	0
EB04	No significant trend	153351	5	2021 - 2025	27.6	0.06	26.78	0.12	0.45

At three program locations, monthly average water temperature increased between 0.05 and 0.06°C per year. No detectable change in monthly average water temperature was observed at one location. There was insufficient data to fit a model for one location.

Salinity - Continuous

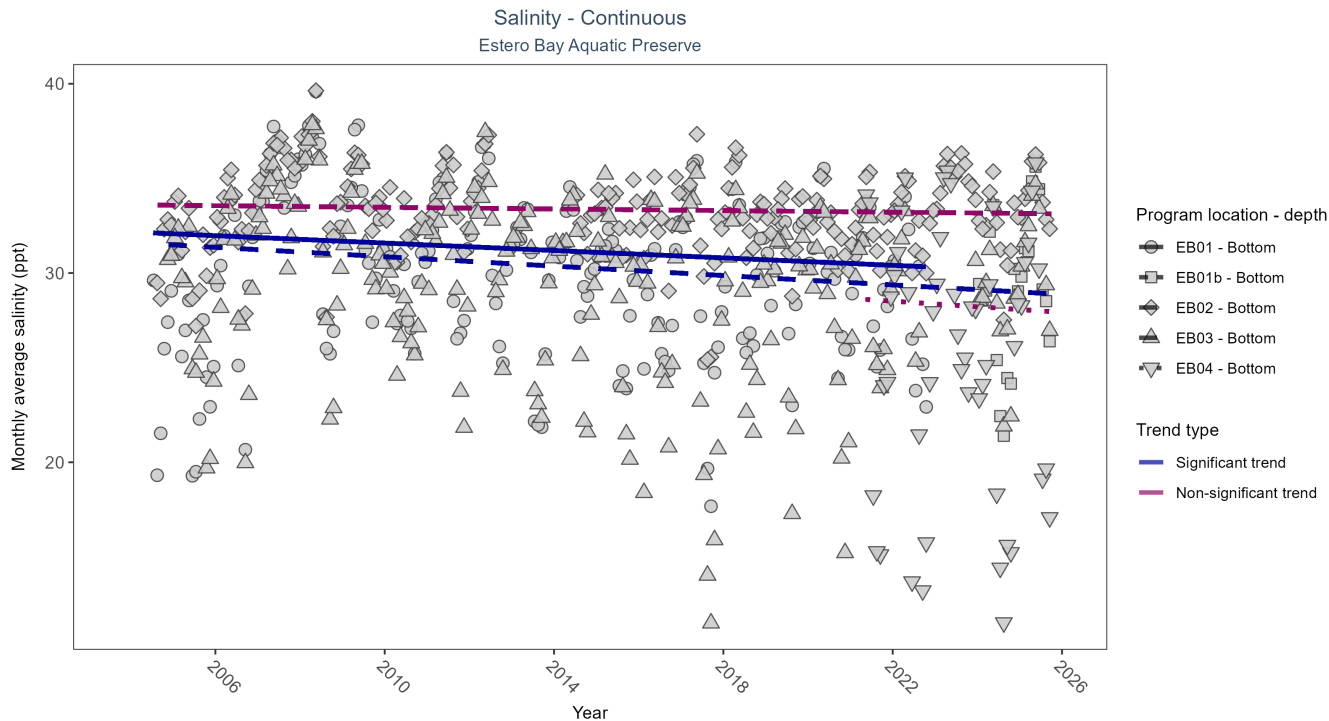


Figure 29: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 34: Seasonal Kendall-Tau Results for Salinity - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
EB01	Significantly decreasing trend	567163	19	2004 - 2022	30.7	-0.13	32.17	-0.1	0.01
EB01b	Insufficient data to calculate trend	56605	2	2024 - 2025	29.8	-	-	-	-
EB02	No significant trend	597294	21	2004 - 2025	33.6	-0.05	33.6	-0.02	0.27
EB03	Significantly decreasing trend	592140	22	2004 - 2025	30.8	-0.19	31.61	-0.12	0
EB04	No significant trend	143333	5	2021 - 2025	27.4	-0.08	28.66	-0.15	0.8

At two program locations, monthly average salinity decreased by 0.1 ppt per year at one site and by 0.12 ppt per year at the other. No detectable change in monthly average salinity was observed at two locations. There was insufficient data to fit a model for one location.

Specific Conductivity - Continuous

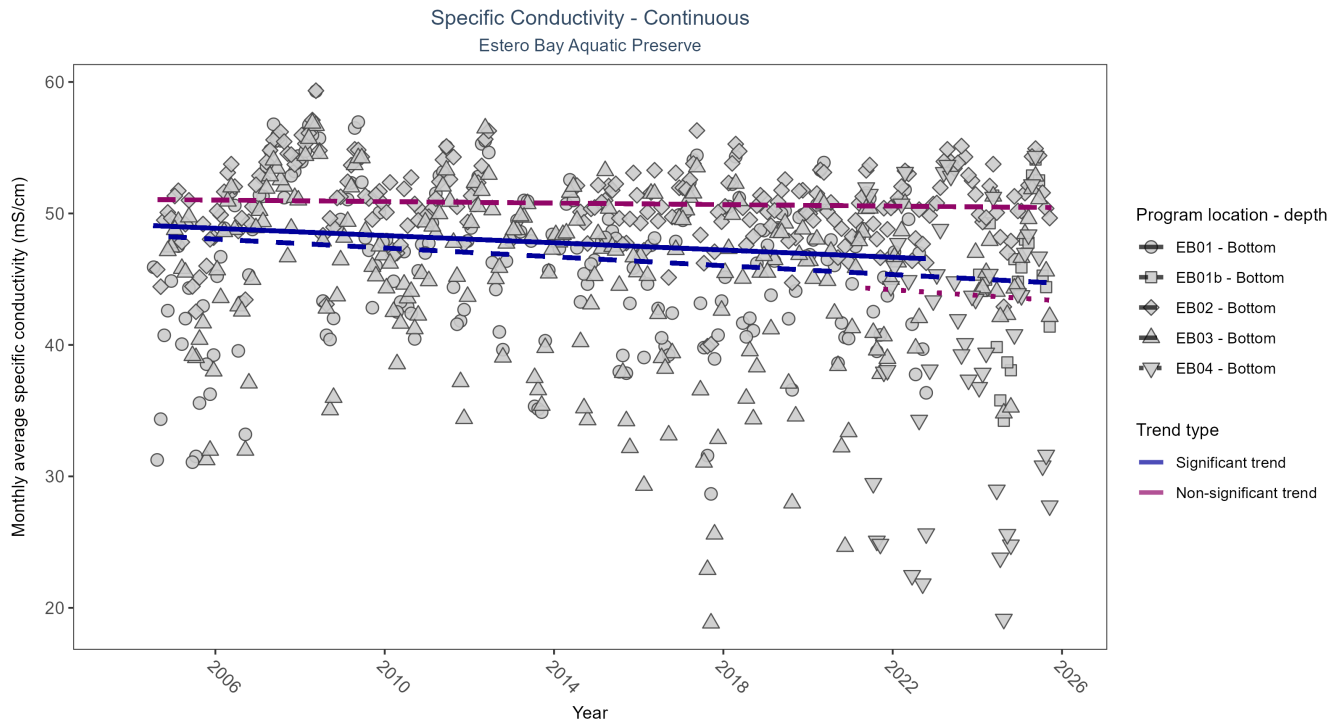


Figure 30: Scatter plot of monthly average specific conductivity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 35: Seasonal Kendall-Tau Results for Specific Conductivity - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
EB01	Significantly decreasing trend	564193	19	2004 - 2022	47.15	-0.13	49.14	-0.14	0.01
EB01b	Insufficient data to calculate trend	56605	2	2024 - 2025	45.92	-	-	-	-
EB02	No significant trend	597301	21	2004 - 2025	51.24	-0.05	51.08	-0.03	0.27
EB03	Significantly decreasing trend	591251	22	2004 - 2025	47.47	-0.19	48.38	-0.17	0
EB04	No significant trend	143333	5	2021 - 2025	42.69	-0.05	44.4	-0.21	0.93

At two program locations, monthly average specific conductivity decreased by 0.14 mS/cm per year at one site and by 0.17 mS/cm per year at the other. No detectable change in monthly average specific conductivity was observed at two locations. There was insufficient data to fit a model for one location.

Submerged Aquatic Vegetation

The data file used is: **All_SAV_Parameters-2026-Jun-04.txt**

Submerged aquatic vegetation (SAV) refers to plants and plant-like macroalgae species that live entirely underwater. The two primary categories of SAV inhabiting Florida estuaries are *benthic macroalgae* and *seagrasses*. They often grow together in dense beds or meadows that carpet the seafloor. *Macroalgae* include multicellular species of green, red and brown algae that often live attached to the substrate by a holdfast. They tend to grow quickly and can tolerate relatively high nutrient levels, making them a threat to seagrasses and other benthic habitats in areas with poor water quality. In contrast, *seagrasses* are grass-like, vascular, flowering plants that are attached to the seafloor by extensive root systems. *Seagrasses* occur throughout the coastal areas of Florida, including protected bays and lagoons as well as deeper offshore waters on the continental shelf. *Seagrasses* have taken advantage of the broad, shallow shelf and clear water to produce two of the most extensive seagrass beds anywhere in continental North America.

Parameters

Percent Cover measures the fraction of an area of seafloor that is covered by SAV, usually estimated by evaluating multiple small areas of seafloor. Percent cover is often estimated for total SAV, individual types of vegetation (seagrass, attached algae, drift algae) and individual species.

Frequency of Occurrence was calculated as the number of times a taxon was observed in a year divided by the number of sampling events, multiplied by 100. Analysis is conducted at the quadrat level and is inclusive of all quadrats (i.e., quadrats evaluated using Braun-Blanquet, modified Braun-Blanquet, and percent cover.”

Species

Turtle grass (*Thalassia testudinum*) is the largest of the Florida seagrasses, with longer, thicker blades and deeper root structures than any of the other seagrasses. It is considered a climax seagrass species.

Shoal grass (*Halodule wrightii*) is an early colonizer of vegetated areas and usually grows in water too shallow for other species except *widgeon grass*. It can often tolerate larger salinity ranges than other seagrass species. *Shoal grass* is characterized by thin, flat blades, that are narrower than *turtle grass* blades.

Manatee grass (*Syringodium filiforme*) is easily recognizable because its leaves are thin and cylindrical instead of the flat, ribbon-like form shared by many other seagrass species. The leaves can grow up to half a meter in length. *Manatee grass* is usually found in mixed seagrass beds or small, dense monospecific patches.

Widgeon grass (*Ruppia maritima*) grows in both fresh and salt water and is widely distributed throughout Florida’s estuaries in less saline areas, particularly in inlets along the east coast. This species resembles *shoal grass* in certain environments but can be identified by the pointed tips of its leaves.

Three species of *Halophila* spp. are found in Florida - **Star grass** (*Halophila engelmannii*), **Paddle grass** (*Halophila decipiens*), and **Johnson’s seagrass** (*Halophila johnsonii*). These are smaller, more fragile seagrasses than other Florida species and are considered ephemeral. They grow along a single long rhizome, with short blades. These species are not well-studied, although surveys are underway to define their ecological roles.

Notes

Star grass, *Paddle grass*, and *Johnson’s seagrass* will be grouped together and listed as **Halophila spp.** in the following managed areas. This is because several surveys did not specify to the species level:

- Banana River Aquatic Preserve
- Indian River-Malabar to Vero Beach Aquatic Preserve
- Indian River-Vero Beach to Ft. Pierce Aquatic Preserve
- Jensen Beach to Jupiter Inlet Aquatic Preserve
- Loxahatchee River-Lake Worth Creek Aquatic Preserve
- Mosquito Lagoon Aquatic Preserve

- Biscayne Bay Aquatic Preserve
- Florida Keys National Marine Sanctuary

Estero Bay Aquatic Preserve
SAV Percent Cover - Sample Locations

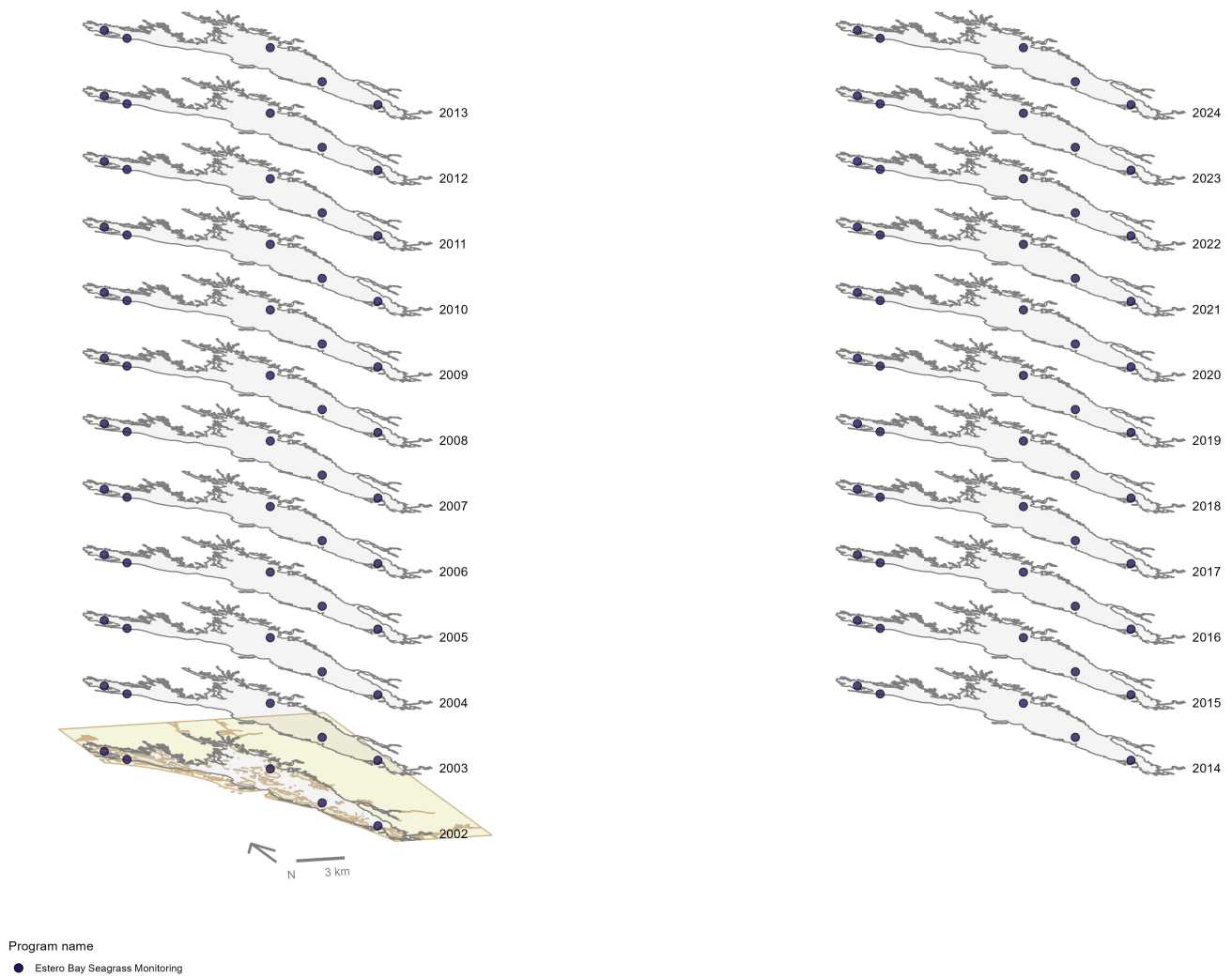


Figure 31: Maps showing the temporal scope of SAV sampling sites within the boundaries of *Estero Bay Aquatic Preserve* by program name.

Click [here](#) to view spatio-temporal plots on GitHub.

Sampling locations by Program:

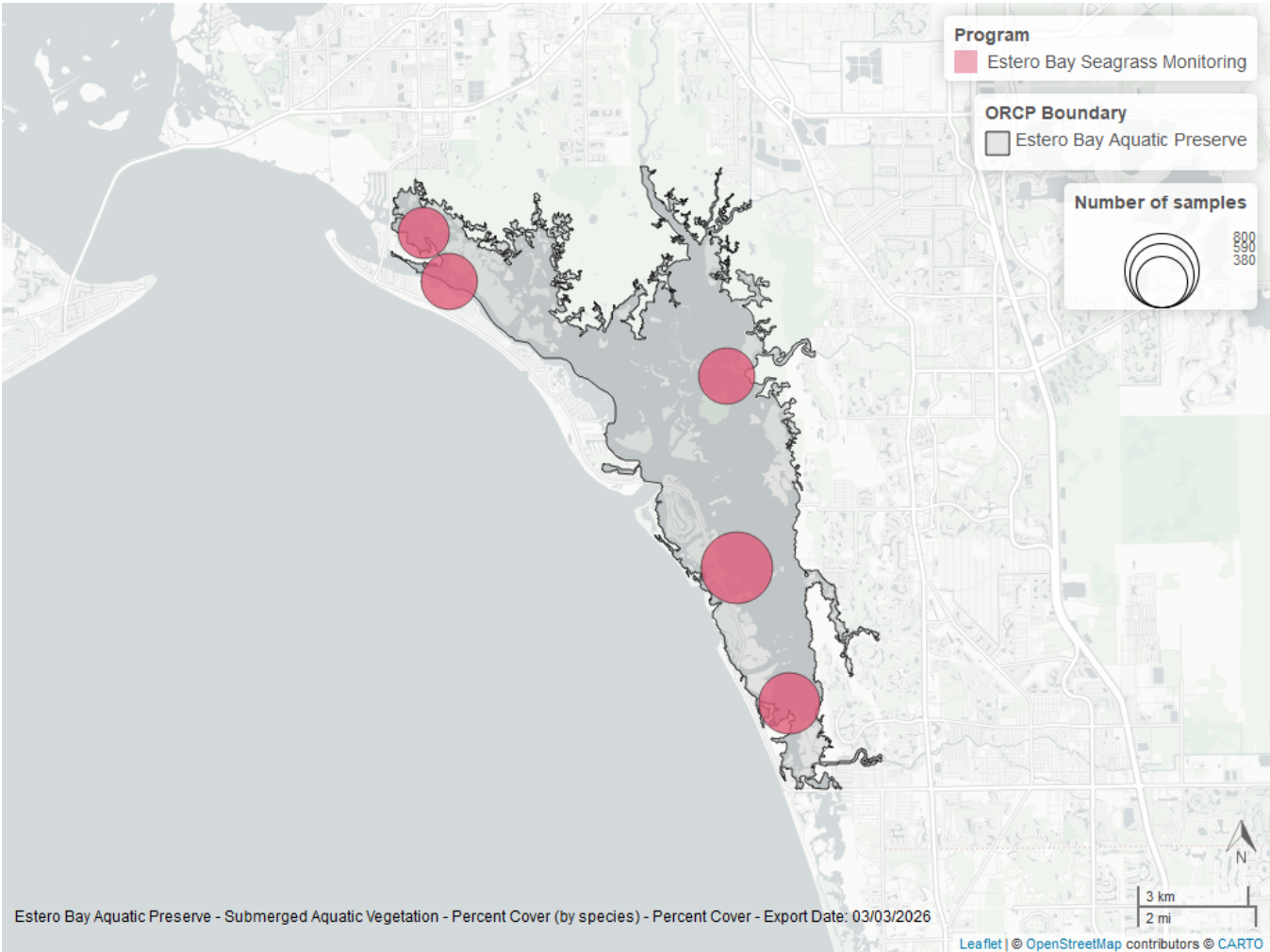


Figure 32: Map showing SAV sampling sites within the boundaries of *Estero Bay Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Table 36: Program Information for Submerged Aquatic Vegetation

<i>ProgramID</i>	<i>N-Data</i>	<i>YearMin</i>	<i>YearMax</i>	<i>method</i>	<i>Sample Locations</i>
571	2715	2002	2024	Braun Blanquet	5

Program names:

[571](#) - Estero Bay Seagrass Monitoring¹⁵

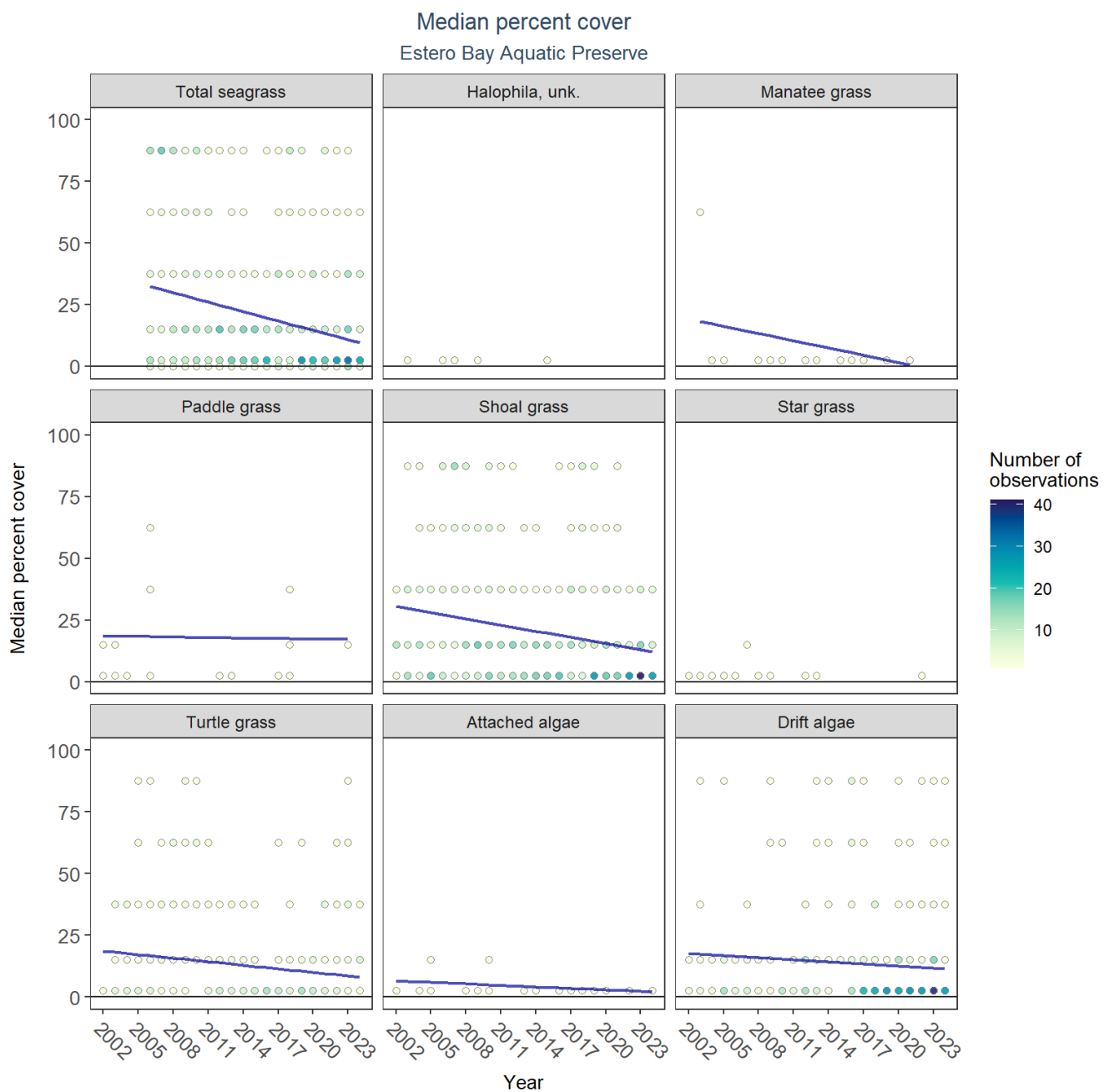


Figure 33: Scatter plots of median percent cover of submerged aquatic vegetation over time by group. Plots for time series that included five or more years of observations show the estimated trend as a blue line.

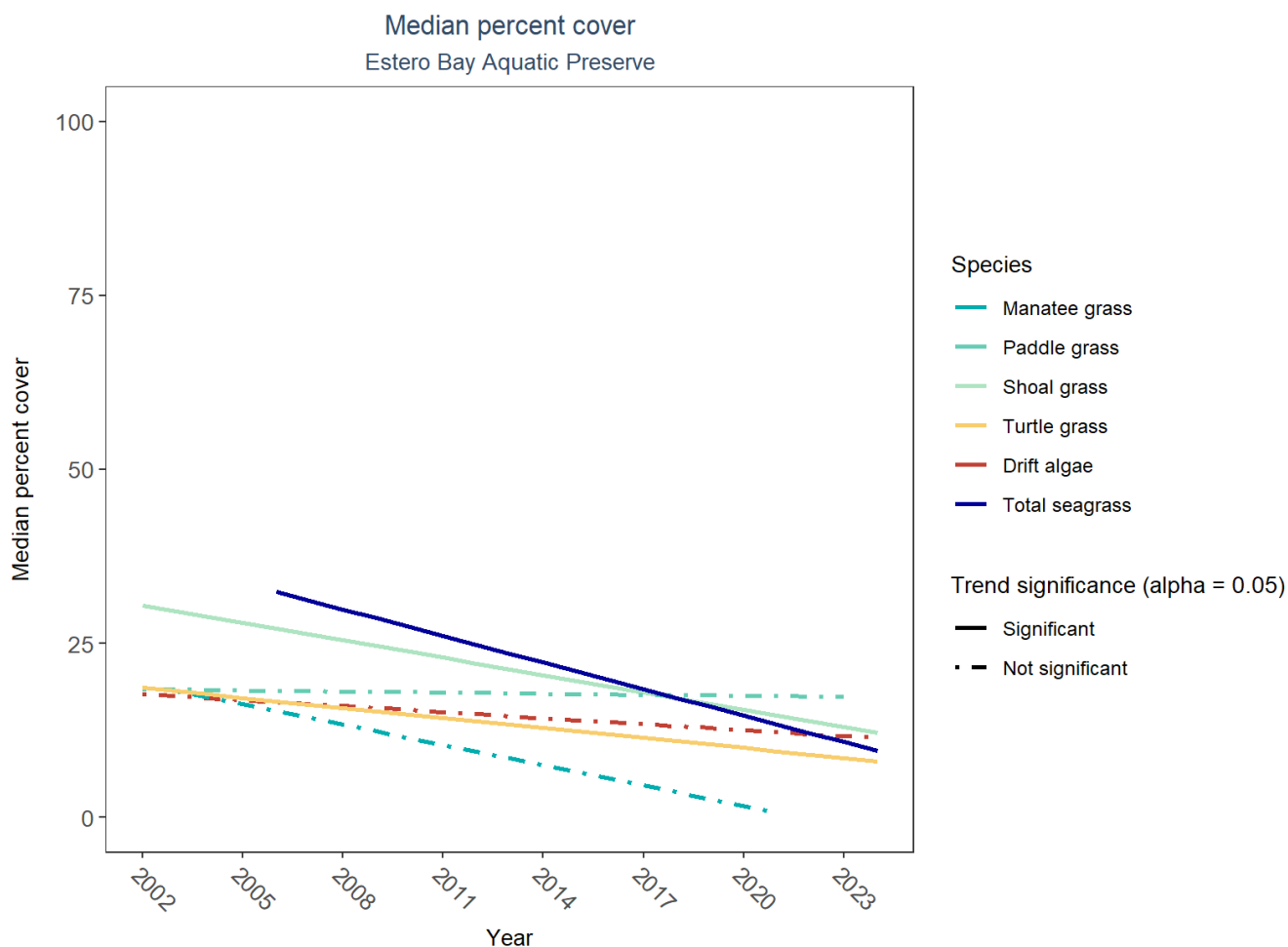


Figure 34: Trend lines of median percent cover of submerged aquatic vegetation over time for species that had five or more years of observations. Line type represents significance (solid) or non-significance (dashed) of the trends.

Table 37: Percent Cover Trend Analysis for Estero Bay Aquatic Preserve

<i>Species</i>	<i>Statistical Trend</i>	<i>Period of Record</i>	<i>LME Intercept</i>	<i>LME Slope</i>	<i>P</i>
Attached algae	No detectable trend	2002 - 2024	8.134705	-0.2019849	0.3861801
Drift algae	No detectable trend	2002 - 2024	20.036336	-0.2876233	0.3785230
Shoal grass	Decreasing trend	2002 - 2024	37.073940	-0.8305677	0.0010585
Paddle grass	No detectable trend	2002 - 2023	18.899337	-0.0551538	0.9221974
Star grass	Model did not fit the available data	2002 - 2022	-	-	-
Manatee grass	No detectable trend	2003 - 2021	26.939075	-0.9721091	0.3517496
Turtle grass	Decreasing trend	2002 - 2024	22.446254	-0.4790471	0.0101184
Total seagrass	Decreasing trend	2006 - 2024	47.664967	-1.2703688	0.0235055
Halophila, unk.	Model did not fit the available data	2003 - 2015	-	-	-

Annual decreases in percent cover were observed for total seagrass (-1.27%), shoal grass (-0.83%), and turtle grass (-0.48%). Manatee grass, paddle grass, attached algae, and drift algae showed no detectable change in percent cover. A model could not be fitted for unknown Halophila and star grass.

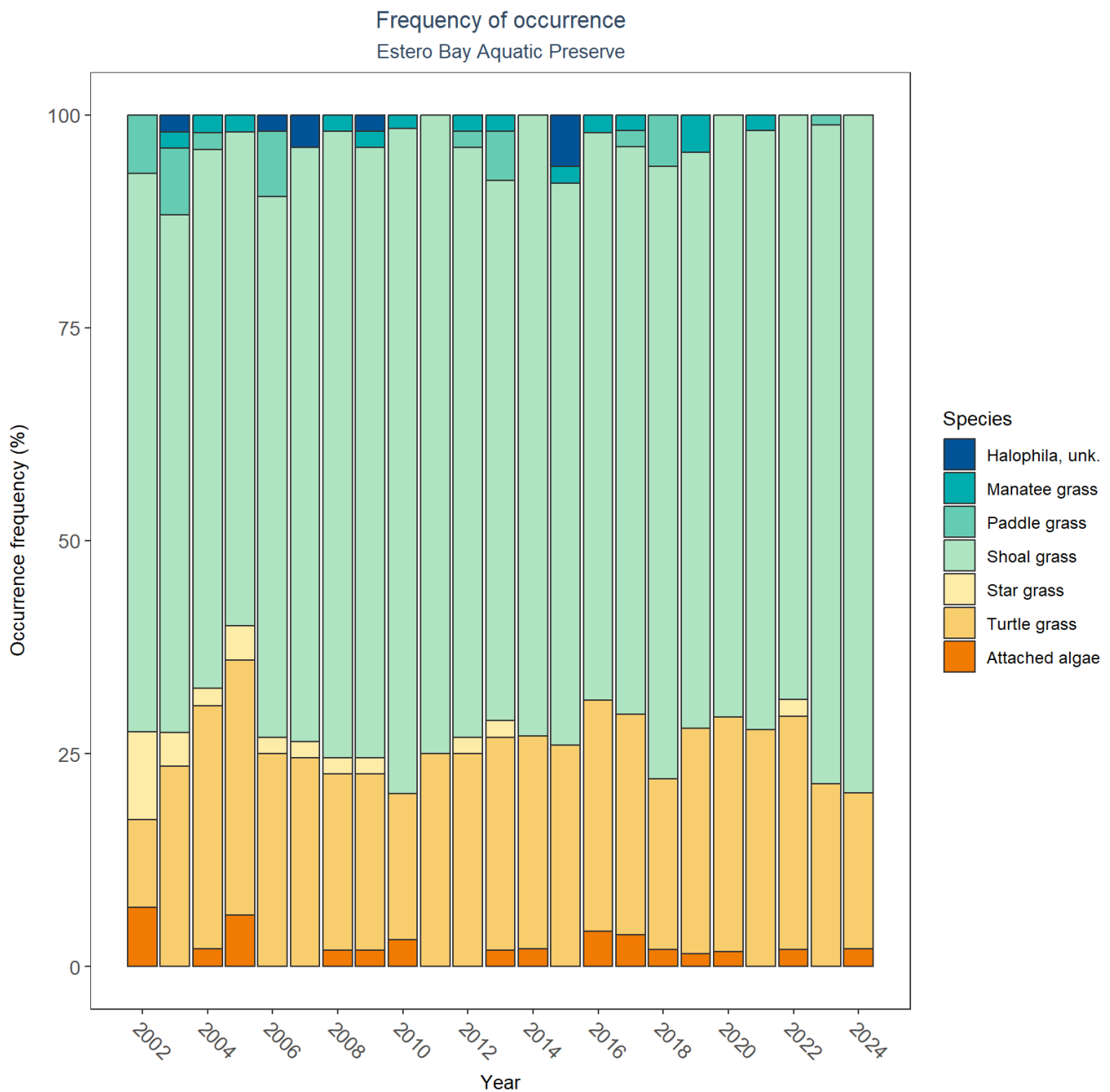


Figure 35: Bar plot of submerged aquatic vegetation species occurrence frequency over time by group.

SAV Water Column Analysis

The following parameters are available for Estero Bay Aquatic Preserve within the SAV_WC_Report:

- Colored Dissolved Organic Matter
- Chlorophyll a
- Dissolved Oxygen
- Dissolved Oxygen Saturation
- pH
- Salinity

- Secchi Depth
- Water Temperature
- Total Nitrogen
- Total Suspended Solids
- Turbidity

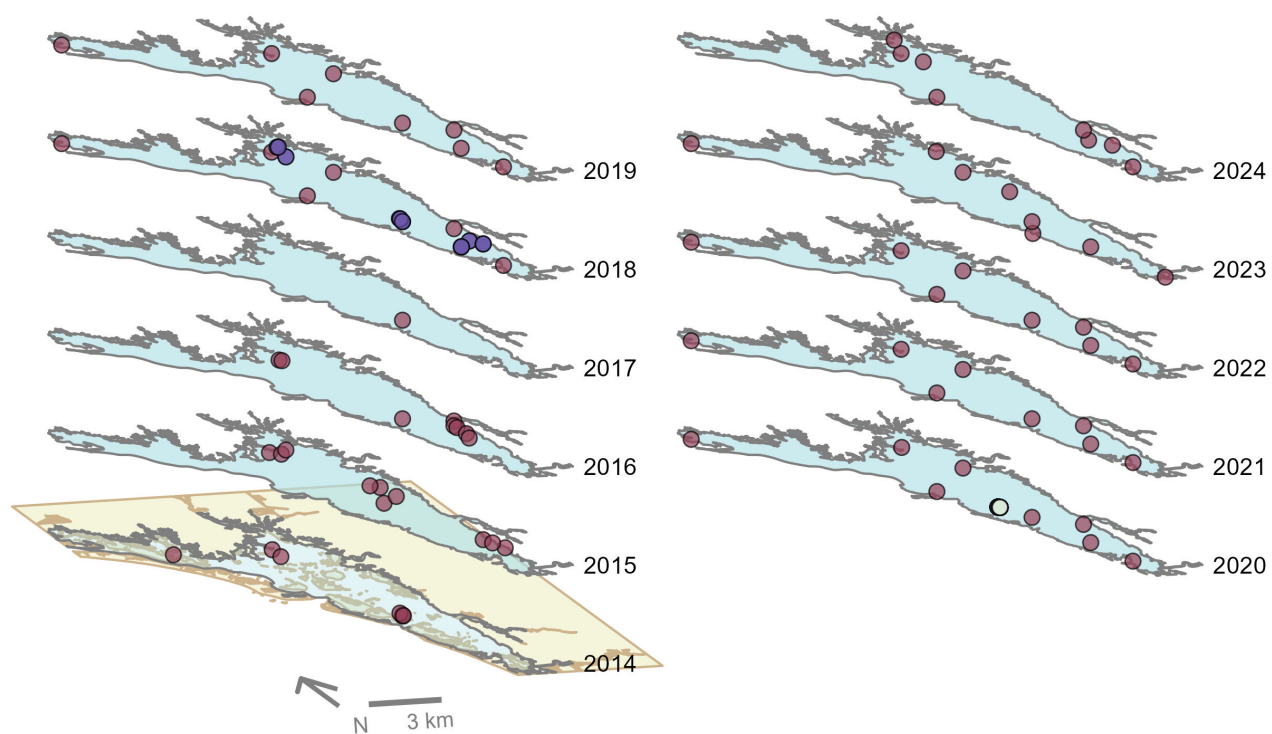
Access the reports here: [DRAFT_SAV_WC_Report.pdf](#)

Oyster

The data file used is: **All_OYSTER_Parameters-2026-Jun-04.txt**

Estero Bay Aquatic Preserve

Oyster sample locations available in SEACAR by Program and Year



Program name

- Estero Bay Oyster Monitoring
- Historical Oyster Body Size (HOBS) Study
- SCCF Oyster Restoration Monitoring

Figure 36: Maps showing the temporal scope of oyster sampling sites within the boundaries of *Estero Bay Aquatic Preserve* by program name.

Click [here](#) to view spatio-temporal plots on GitHub.

Density

For natural reefs, density showed no detectable trend between 2014 and 2024. For restored reefs, there was insufficient data to calculate a trend for density.

Restored

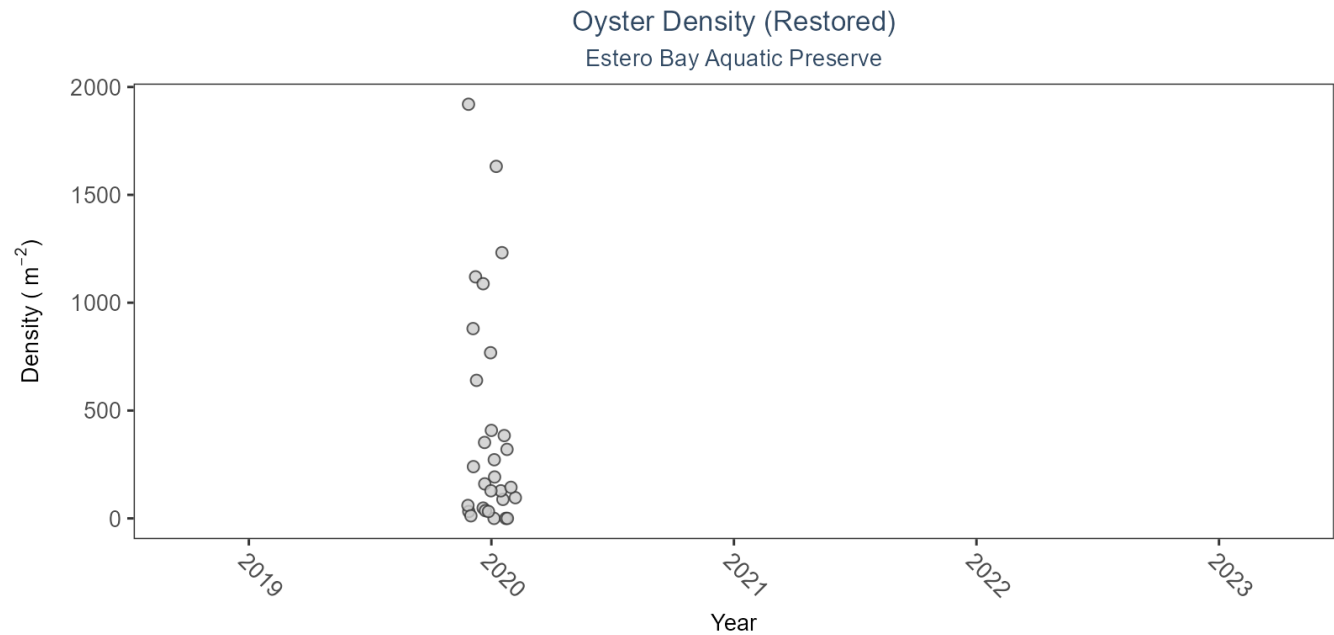


Figure 37: Scatter plot of oyster density over time. If the time series included five or more years with observations, an estimated trend (blue line) and a 95% credible interval (purple band) may also be plotted. Data points are jittered horizontally to reduce overlap.

Table 38: Model results for Oyster Density - Restored

<i>Habitat Type</i>	<i>Shell Type</i>	<i>Statistical Trend</i>	<i>Period of Record</i>	<i>Estimate</i>	<i>Standard Error</i>	<i>Credible Interval</i>
Restored	Live Oysters	Insufficient data	2020 - 2020	-	-	-

Natural

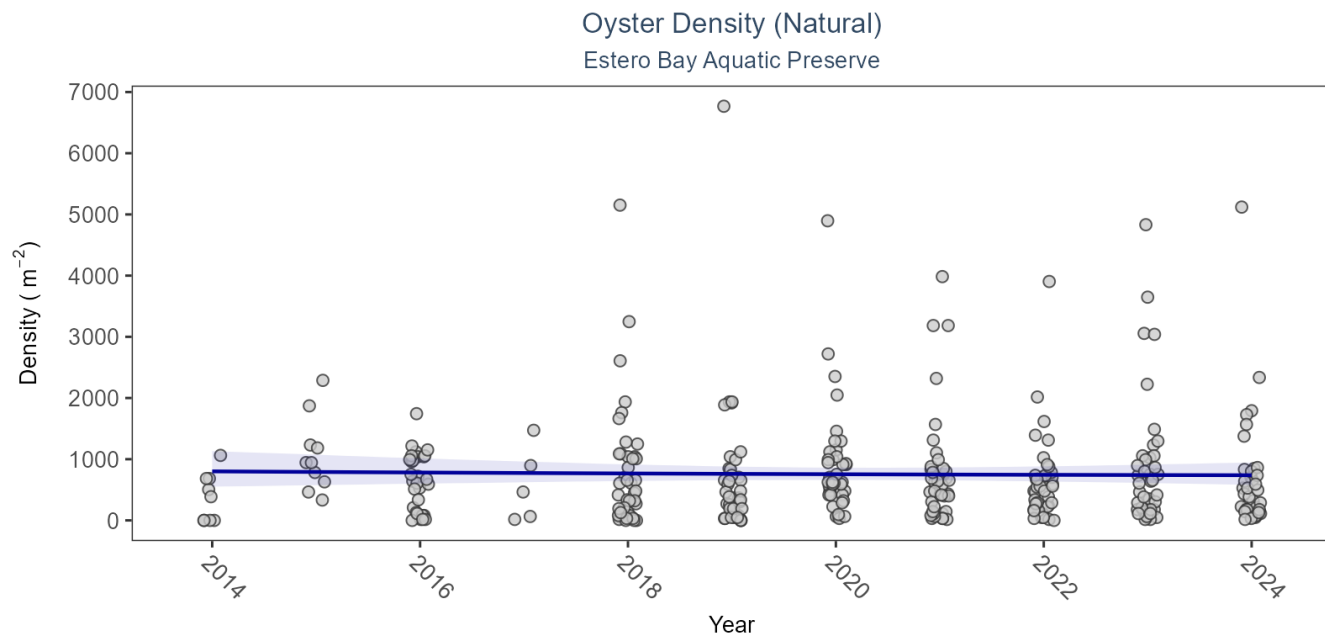


Figure 38: Scatter plot of oyster density over time. If the time series included five or more years with observations, an estimated trend (blue line) and a 95% credible interval (purple band) may also be plotted. Data points are jittered horizontally to reduce overlap.

Table 39: Model results for Oyster Density - Natural

<i>Habitat Type</i>	<i>Shell Type</i>	<i>Statistical Trend</i>	<i>Period of Record</i>	<i>Estimate</i>	<i>Standard Error</i>	<i>Credible Interval</i>
Natural	Live Oysters	No detectable trend	2014 - 2024	-6.31	21.6	-50.66 to 34.7

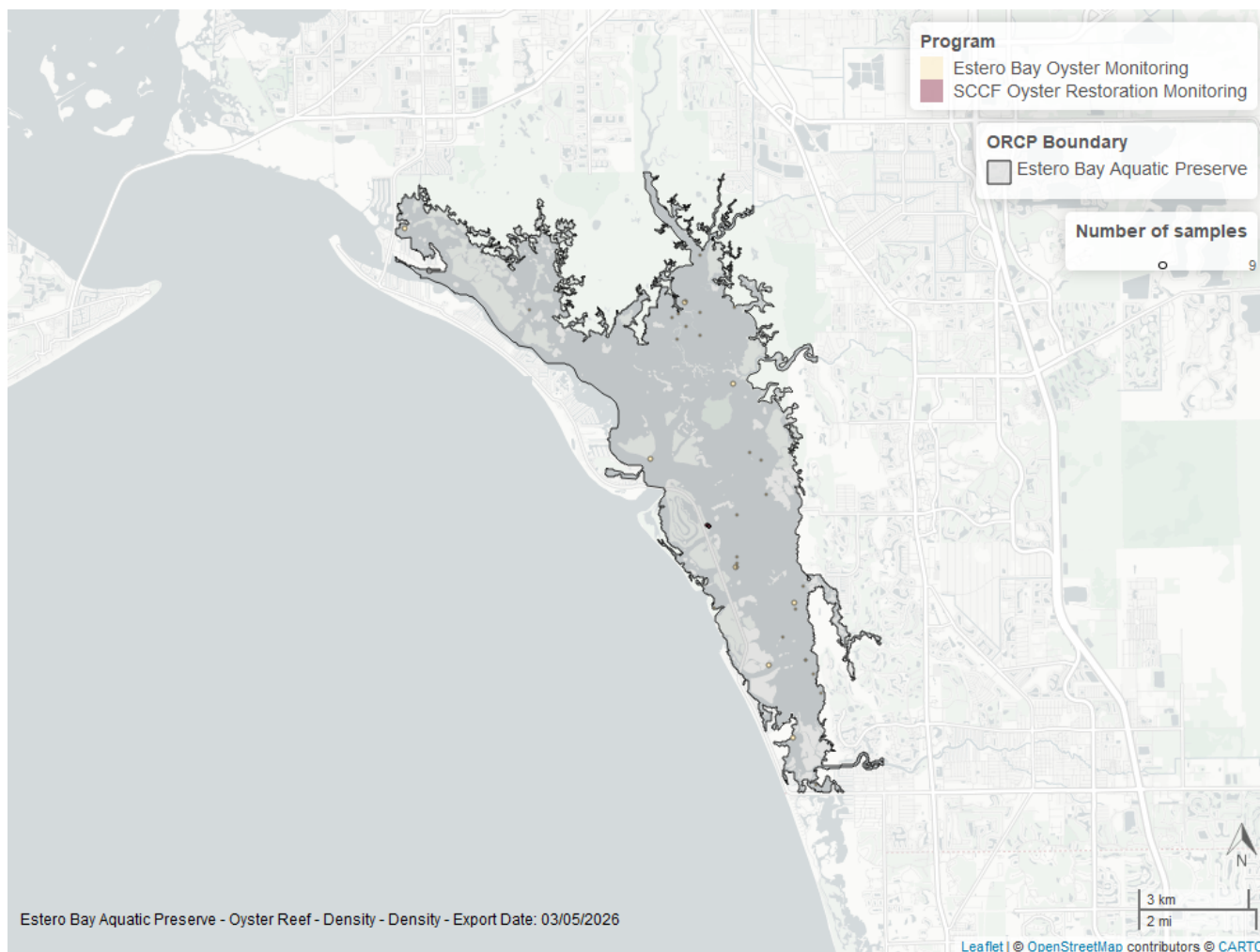


Figure 39: Map showing location of oyster density sampling locations within the boundaries of *Estero Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Percent Live

For natural reefs, percent live cover showed no detectable trend between 2016 and 2024. For restored reefs, there was insufficient data to calculate a trend for percent live cover.

Restored

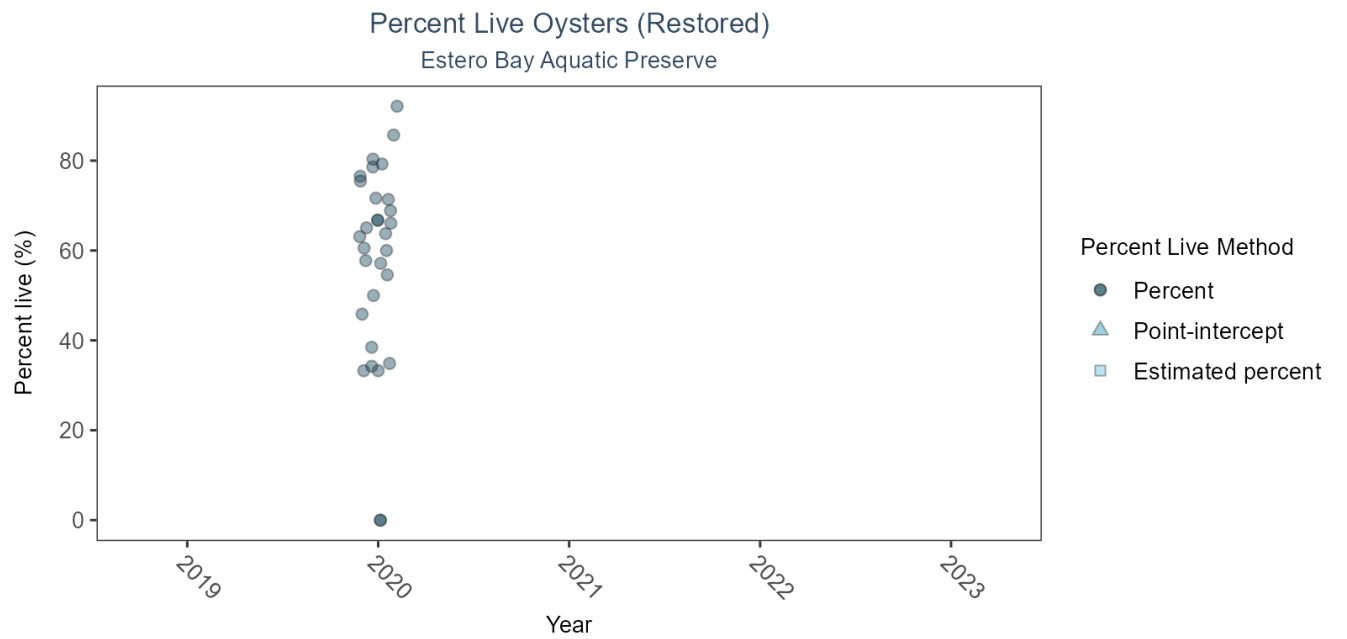


Figure 40: Scatter plot of percent live oysters over time. If the time series included five or more years with observations, an estimated trend (blue line) and a 95% credible interval (purple band) may also be plotted. Estimation method is represented as percent (circles), point-intercept (triangle), or estimated percent (square), with data points jittered horizontally to reduce overlap.

Table 40: Model results for Oyster Percent Live - Restored

<i>Habitat Type</i>	<i>Shell Type</i>	<i>Statistical Trend</i>	<i>Period of Record</i>	<i>Estimate</i>	<i>Standard Error</i>	<i>Credible Interval</i>
Restored	Live Oysters	Insufficient data	2020 - 2020	-	-	-

Natural

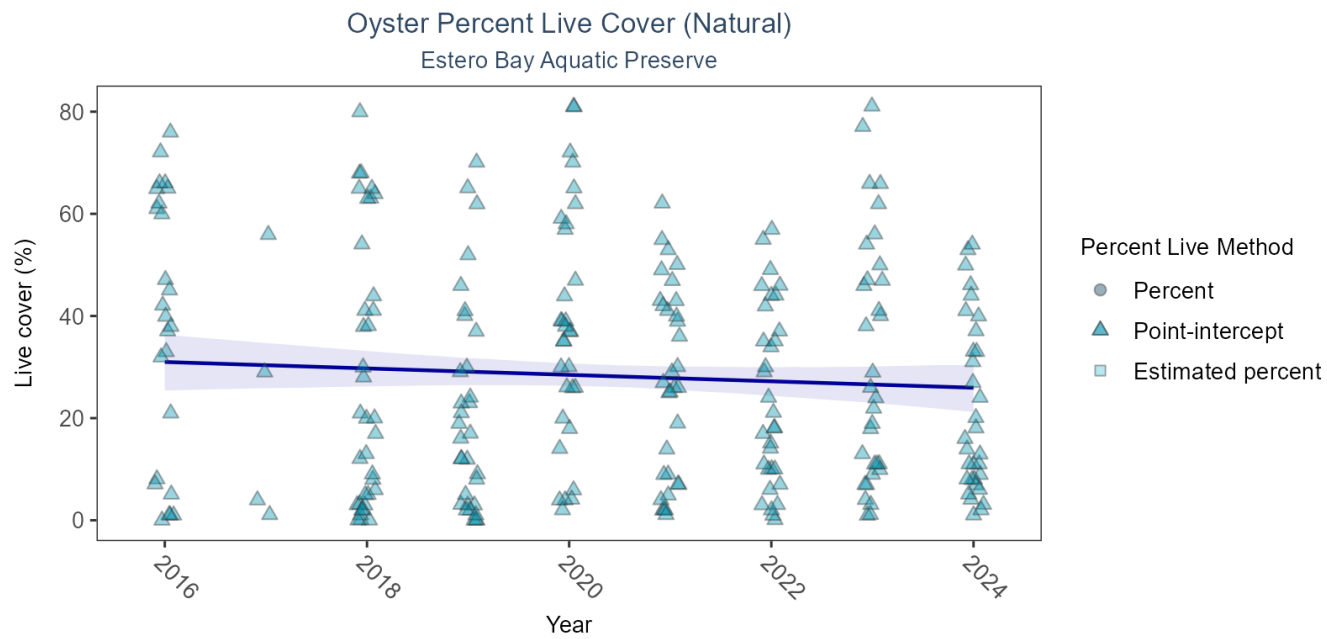


Figure 41: Scatter plot of percent live oysters over time. If the time series included five or more years with observations, an estimated trend (blue line) and a 95% credible interval (purple band) may also be plotted. Estimation method is represented as percent (circles), point-intercept (triangle), or estimated percent (square), with data points jittered horizontally to reduce overlap.

Table 41: Model results for Oyster Percent Live - Natural

<i>Habitat Type</i>	<i>Shell Type</i>	<i>Statistical Trend</i>	<i>Period of Record</i>	<i>Estimate</i>	<i>Standard Error</i>	<i>Credible Interval</i>
Natural	Live Oysters	No detectable trend	2016 - 2024	-0.63	0.58	-1.83 to 0.49

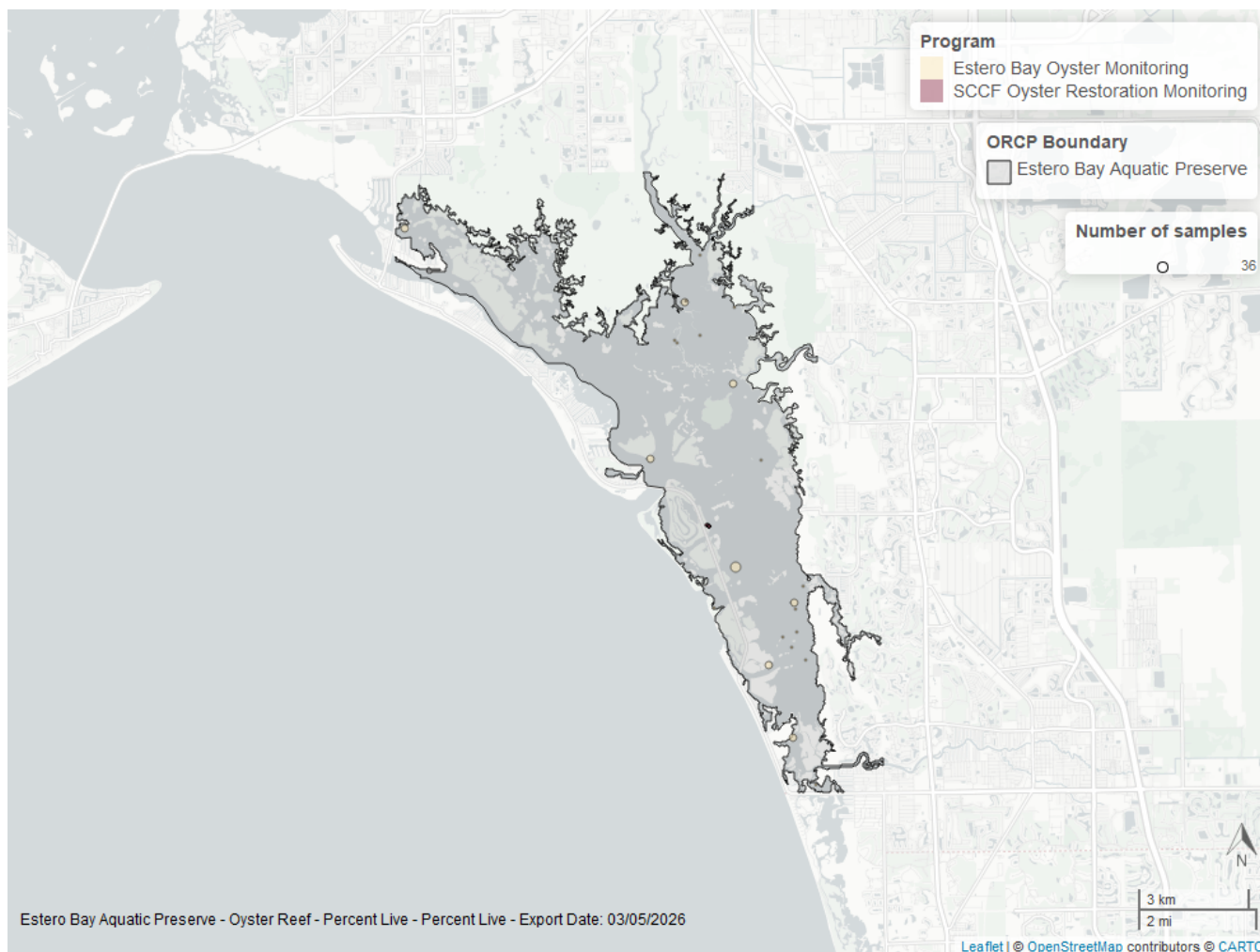


Figure 42: Map showing location of oyster percent live sampling locations within the boundaries of *Estero Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Shell Height

For natural reefs, annual average live oyster shell height in the 25-75mm size class decreased by 0.42mm per year, and there was no detectable trend for live oysters in the ≥ 75 mm size class. For restored reefs, there was insufficient data to calculate a trend for live oysters in either the 25-75mm or the ≥ 75 mm size class. Models are not run on dead oyster shell measurements.

Restored

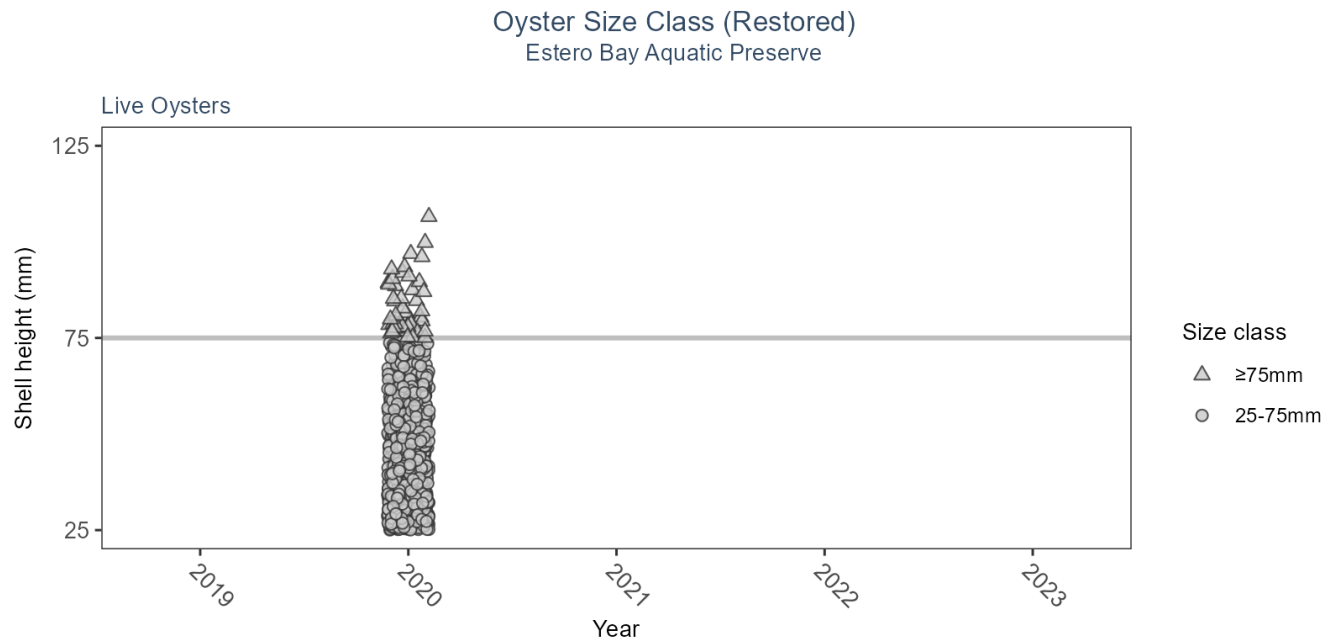


Table 42: Model results for Oyster Shell Height - Restored

Habitat Type	Shell Type	SizeClass	Statistical Trend	Period of Record	No. of Samples	Estimate	Standard Error	Credible Interval
Restored	Live Oysters	>75mm	Insufficient data	2020 - 2020	47	-	-	-
Restored	Live Oysters	25-75mm	Insufficient data	2020 - 2020	824	-	-	-

Natural

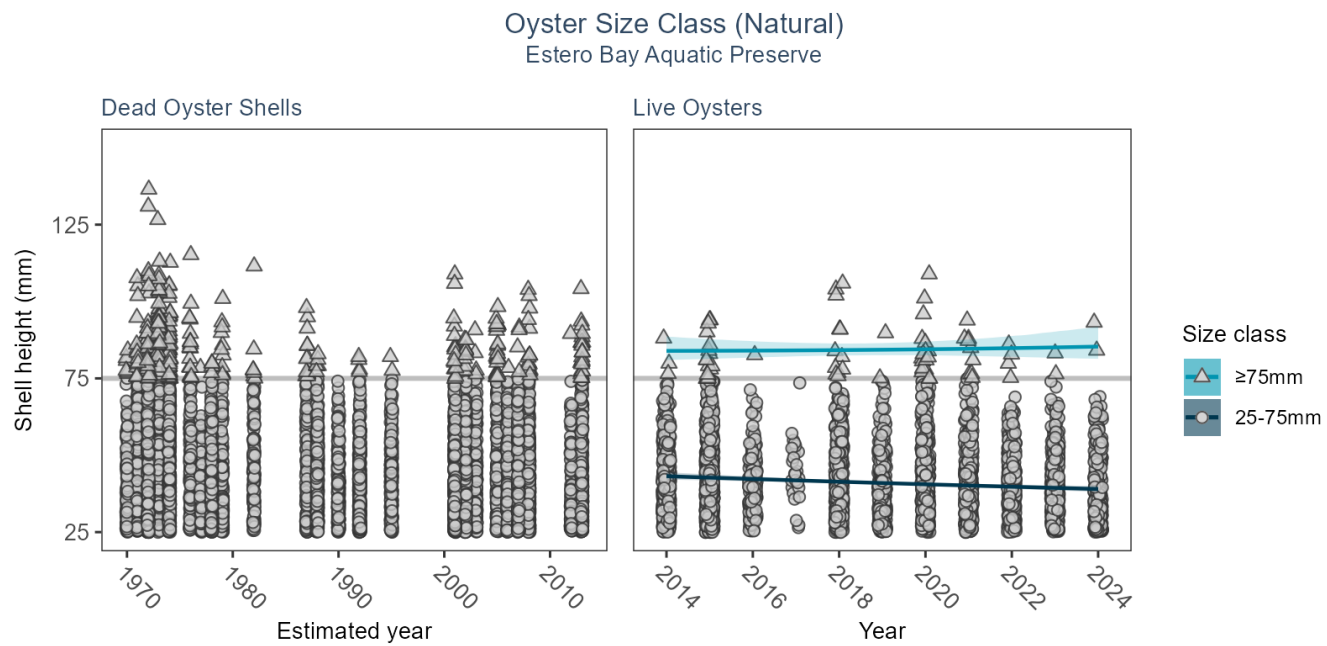


Table 43: Model results for Oyster Shell Height - Natural

<i>Habitat Type</i>	<i>Shell Type</i>	<i>SizeClass</i>	<i>Statistical Trend</i>	<i>Period of Record</i>	<i>No. of Samples</i>	<i>Estimate</i>	<i>Standard Error</i>	<i>Credible Interval</i>
Natural	Dead Oyster Shells	>75mm	Model not run on dead oyster shell	1970 - 2013	332	-	-	-
Natural	Dead Oyster Shells	25-75mm	Model not run on dead oyster shell	1970 - 2013	6086	-	-	-
Natural	Live Oysters	>75mm	No detectable trend	2014 - 2024	70	0.14	0.4	-0.62 to 0.96
Natural	Live Oysters	25-75mm	Decreasing trend	2014 - 2024	3561	-0.42	0.1	-0.63 to -0.22

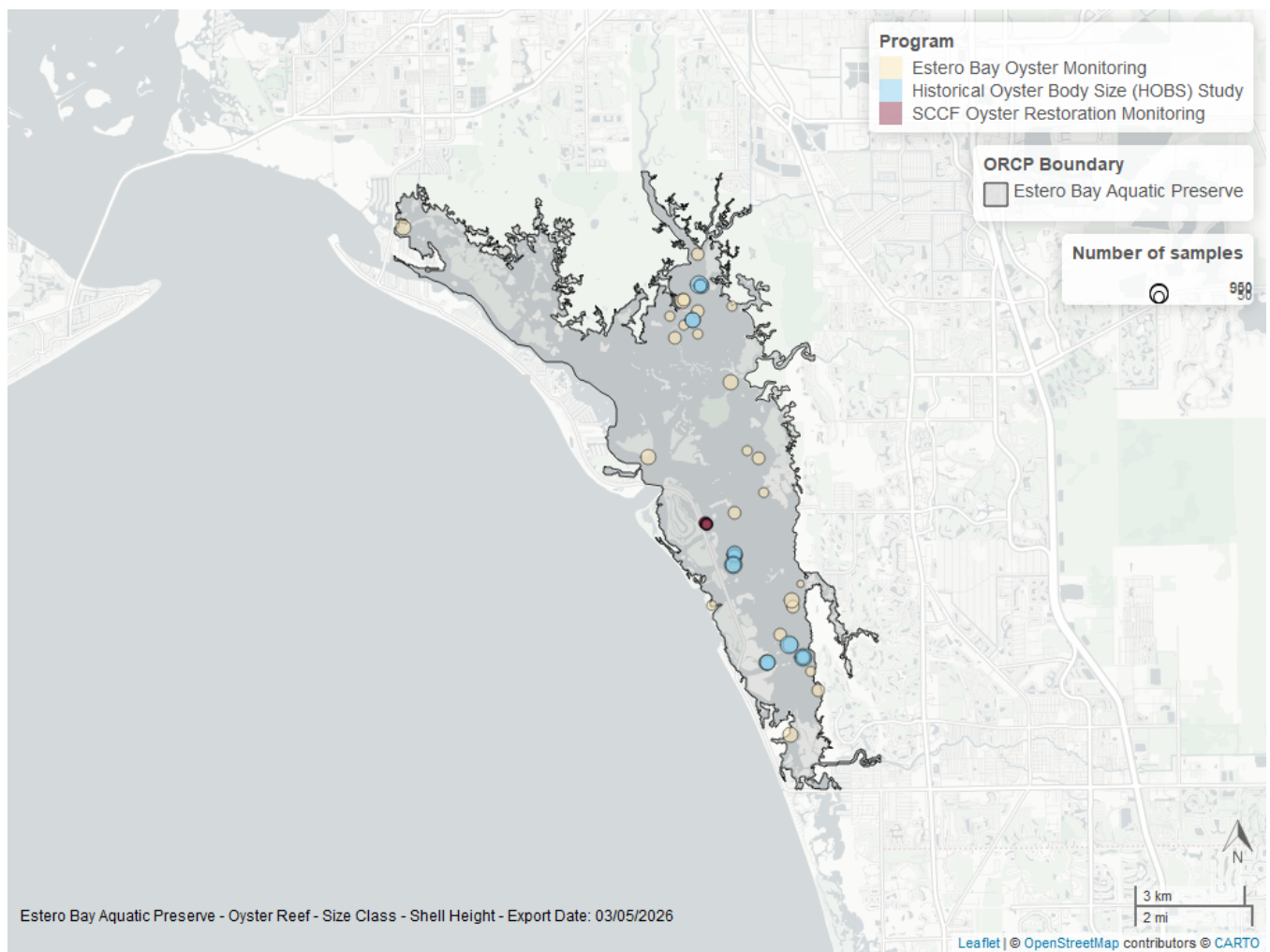


Figure 43: Map showing location of oyster shell height sampling locations within the boundaries of *Estero Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Species list

Caulerpa prolifera ¹	Halodule wrightii ¹	No grass in quadrat ¹
Caulerpa sertularioides ¹	Halophila decipiens ¹	Syringodium filiforme ¹
Caulerpa spp. ¹	Halophila engelmannii ¹	Thalassia testudinum ¹
Drift algae ¹	Halophila sp. ¹	Total seagrass ¹

1 - Submerged Aquatic Vegetation

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